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Research article

Electroencephalogram-based emotion recognition: a comparative analysis of supervised machine learning algorithms

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ABSTRACT

Emotion recognition from electroencephalogram (EEG) signals has garnered significant attention owing to its potential applications in affective computing, human-computer interaction, and mental health monitoring. This paper presents a comparative analysis of different machine learning methods for emotion recognition using EEG data. The objective of this study was to identify the most effective algorithm for accurately classifying emotional states using EEG signals. The EEG brainwave dataset: Feeling emotions dataset was used to evaluate the performance of various machine-learning techniques. Multiple machine learning techniques, namely logistic regression (LR), support vector machine (SVM), Gaussian Naive Bayes (GNB), and decision tree (DT), and ensemble models, namely random forest (RF), AdaBoost, LightGBM, XGBoost, and CatBoost, were trained and evaluated. Five-fold cross-validation and dimension reduction techniques, such as principal component analysis, *t*-distributed stochastic neighbor embedding, and linear discriminant analysis, were performed for all models. The least-performing model, GNB, showed substantially increased performance after dimension reduction. Performance metrics such as accuracy, precision, recall, F1-score, and receiver operating characteristic curves are employed to assess the effectiveness of each approach. This study focuses on the implications of using various machine learning algorithms for EEG-based emotion recognition. This pursuit can improve our understanding of emotions and their underlying neural mechanisms.

1. Introduction

Emotions are an inherent part of human life. They play a pivotal role in shaping decisions, extending beyond actual choices, thereby directly impacting our daily lives. Emotion recognition models help to understand and respond to human emotions and find applications for inspecting driver emotions (Gamage et al., 2022), healthcare, entertainment, and software programs for user satisfaction (Geetha et al., 2024). Emotion recognition models broadly follow two approaches for recognizing emotions: behavioral and physiological. Behavioral modalities include emotion recognition using facial expressions (Kim et al., 2021), gestures (Wei et al., 2024), and speech (Chen et al., 2024; Khan et al., 2024). In contrast, physiological modalities include emotion recognition from physiological signals such as electroencephalogram (EEG), electrocardiography (ECG), galvanic skin response (GSR), and electrodermal activity (EDA) (Saganowski et al., 2023; Shu et al., 2018). The subject can control these behavioral modalities, and the quality of their expression may be significantly influenced by their personality and current environment. However, physiological signals are continuously available and cannot be intentionally or consciously controlled. Integrating both modalities can provide a more comprehensive understanding of an individual's emotional state.

Most studies on emotion recognition have analyzed how participants responded to controlled stimuli in a laboratory setting. Consequently, effective identification of emotions in real-world scenarios remains a significant challenge. Applying automatic emotion recognition through physiological signals captured by wearable devices can drive a revolutionary transformation in healthcare, human-computer interfaces, automotive technology, gaming, and e-learning (Pal et al., 2021; Yamin et al., 2023). A breakthrough in this can revolutionize the identification of mental health and can address concerns, such as depression and anxiety disorders, thereby significantly improving overall well-being. Individuals

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with Autism Spectrum Disorder (Mayor Torres et al., 2022) could benefit from personal assistants or wearables that aid in perceiving and expressing emotions. Moreover, incorporating emotional awareness can enhance existing systems, improve recommendations, user interfaces, and content relevance, as well as fix game difficulty and emotionally attuned vehicles. Furthermore, studies on emotion recognition can provide crucial support to emotionally fragile individuals dealing with past trauma, grief, and emotional imbalances by offering real-time monitoring of their emotional states and providing insights into emotional triggers. This technology enhances therapeutic effectiveness, promotes self-regulation, fosters greater emotional awareness, and simultaneously improves mental health.

Anatomically speaking, the human brain's cerebral cortex predominantly controls cognitive and higher-order emotional functions. The limbic system plays a crucial role in emotional processing, and components such as the amygdala and hippocampus are involved in emotional responses and memory formation. The brain functions via the transmission of electrical signals between neurons. One approach to studying the electrical activity of the brain is to noninvasively capture the potential resulting from brain activity from the scalp. The recorded signal representing potential variations between two points is called an EEG. Thus, EEG can be used to identify the brain regions closely related to emotions.

EEG signals allow for a more comprehensive understanding of emotional states that would have otherwise been hidden (Stajić et al., 2022). In EEG, electrodes placed on a subject's scalp detect electrical activity primarily generated by groups of pyramidal neurons in the cerebral cortex (Britton et al., 2016). The electrical activity recorded on the scalp represents a combination of inhibitory or excitatory postsynaptic potentials from multiple pyramidal cells near each electrode. It can be used to measure brain activity in the presence or absence of certain events or stimuli, making it suitable for emotion classification. Using brain signals to distinguish emotions has the potential to transform the identification and management of a wide range of health issues, including stroke, stress management, epilepsy, autism spectrum disorder, and other anxiety disorders. Utilizing EEG for emotion recognition has the potential to provide a deeper understanding of the emotional states of people with a variety of neurodiverse ailments.

This study is grounded in an interdisciplinary theoretical framework combining psychological, neurophysiological, and computational models for emotion recognition. Central to this framework is Lövheim's three-dimensional emotional model (Lövheim, 2012), which links the brain's chemical composition to generalized emotional states characterized by varying levels of positive and negative valence through monoamine neurotransmitters such as dopamine, norepinephrine, and serotonin. Lövheim's model provides a biological basis for understanding how physiological markers relate to emotional states, supporting the use of EEG signals as measurable indicators of internal states. This is complemented by neurophysiological theories highlighting the role of brain regions, such as the amygdala and cerebral cortex, in emotion regulation and processing. Furthermore, this study applies principles from machine-learning theory, especially supervised learning, to identify patterns within EEG data that correlate with specific emotional states. Together, these theoretical foundations underscore our dual-modal approach to emotion recognition, addressing gaps in real-world applications and setting the stage for transformative applications in healthcare, human-computer interaction, and adaptive technology, which can dynamically respond to users' emotional needs.

Although EEG-based emotion recognition allows measurements on unbiased terms, the research-grade EEG setup has a complex installation, and equipment maintenance is prone to movement artifacts. However, with the increased availability of commercial EEG devices, emotion recognition has gained attraction (Dadebayev et al., 2022). This study analyzed data in the frequency domain of the EEG sensor channels. Removing artifacts from EEG signals, extracting spectral information from the frequency domain of the EEG signal, and devising a multiclass classification technique are steps in detecting emotions using EEG signals. The main contributions of this study are as follows.

- We present a comparative analysis of different machine learning models for emotion recognition using EEG data. For comparative analysis, we used machine models, such as logistic regression (LR), support vector machine (SVM), Gaussian Naive Bayes (GNB), k-nearest neighbor (KNN), decision tree (DT), and ensemble models, namely random forest (RF), AdaBoost, LightGBM, XGBoost, and CatBoost.
- The performance of the machine learning models based on metrics such as accuracy, precision, recall, F1-score, and receiver operating characteristic (ROC) curves were analyzed. Our comparative analysis demonstrated the significance of each machine learning model for emotion recognition using the EEG dataset.
- Each machine learning model's hyperparameter tuning and computational time and estimated the optimum hyperparameters for accurate emotion recognition were considered. The significance of the hyperparameters for emotion recognition was assessed by considering the hyperparameters of each model with different values. The computational times before and after hyperparameter tuning were calculated, and the model performance was evaluated based on computational complexity.
- We evaluated the significance of the k-fold cross-validation approach for machine and ensemble models based on cross-validation scores and classification accuracy. We tested each model with fold changes ranging from 1 to 5 and analyzed its classification performance. The k-fold cross-validation analysis summarizes the emotion recognition performance of the machine and ensemble models when the dataset is divided into k subsets or folds.
- We implemented a dimension-reduction algorithm for lower-performing machine learning models and analyzed its classification performance for emotion recognition. The implementation of dimension-reduction approaches significantly improved the classification performance of lower-performing machine models.
- This study tested the effects of the feature selection technique for emotion recognition. Feature selection algorithms identify significant features that influence a model's performance. Feature selection techniques find the best set of features that allow one to build optimized machine-learning models.

From a management practice perspective, our methodology introduces machine learning models for emotion classification, which have practical implications for enhancing user-centered applications across industries. For example, healthcare management can benefit from tools for monitoring mental health conditions, such as depression and anxiety, in real time, allowing for early intervention and improved patient care. Similarly, emotion recognition insights can inform management in fields such as customer service and human resources, where emotional feedback is critical. This study's comparative analysis of machine learning models further addresses the gap in the literature concerning model performance, interpretability, and computational efficiency, offering guidance for selecting and tuning algorithms based on specific industrial needs. By implementing the hyperparameters tuning, k-fold cross-validation, and feature selection techniques, this study provides a scalable and optimized approach for building reliable emotion recognition systems that are adaptable to various practical applications in management, including user experience optimization, mental health monitoring, and adaptive interface design.

The remainder of this paper is organized as follows: Section 2 presents related work on emotion recognition using EEG signals. Section 3 provides a system overview of emotion recognition, and the various supervised machine-learning models used in our study. Section 4 describes the experimental results and analysis of supervised machine learning-based

emotion recognition systems. The limitations and future research directions of EEG-based emotion recognition are presented in Section 5. Finally, we present the conclusions in Section 6.

2. Related work

An overview of the literature on EEG-based emotion recognition shows the development of this discipline, with studies addressing various topics, including signal processing, feature extraction, classification methods, and applications (Bota et al., 2019; Egger et al., 2019; Lin and Li, 2023). Our literature review sheds light on the development of EEG-based emotion recognition and emphasizes new directions, obstacles, and achievements in this multidisciplinary field. Researchers have investigated novel methodologies to enhance the precision, resilience, and practicality of EEG-based emotion identification systems (Ahmad and Khan, 2022; Imani and Ali Montazer, 2019).

Early research on EEG-based emotion detection aimed to determine whether emotional information can be extracted from brainwaves. The connection between passionate emotions and brain activity, particularly in frontal regions, has been the subject of numerous research (Barros et al., 2022). Frontal alpha asymmetry has been investigated in the context of emotional processing, in which disparities in alpha brainwave activity in the frontal cortex are associated with various emotional states. In Allen and Reznik (2015), frontal EEG asymmetry as a promising marker of depression vulnerability is studied. It may be possible to identify those who are at risk for depression due to frontal asymmetry; however, a large-scale prospective investigation is required. Furthermore, research in this field can guide the development of neurally informed, individualized treatments and enable an evaluation of how well treatments (including psychotherapy) alter the activity within these neural systems. This is because frontal asymmetry may index the activity of neural systems that promote the risk of depression. In Mennella et al. (2017), frontal alpha asymmetry neurofeedback for the alleviation of anxiety and negative affect was described. In this study, they used neurofeedback training to assess discrete changes in alpha power at the left and right locations as well as in positive and negative affect, anxiety, and depression. These groundbreaking studies laid the foundation for subsequent investigations into the neural correlates of emotions using EEG.

Various studies have explored signal processing methods to improve the quality of EEG data for emotion identification (Chaddad et al., 2023; Negi, 2021; Petrantonakis and Hadjileontiadis, 2015). Studies have examined various methods to improve the quality of EEG signals by eliminating artifacts, including eye blinks, muscular activity, and background noise. Researchers have investigated cutting-edge methods for denoising and filtering to increase signal-to-noise ratios. Li et al. (2021) proposed a modified independent component analysis (ICA) approach to remove myoelectric artifacts from EEG signals. In this method, the greatest power from electromyogram (EMG) artifacts was "forced" into a small number of independent components by adding extra channels of genuine EMG from the neck and head muscles (reference artifacts) as inputs to ICA. Their proposed method achieved a more precise EMG separation without manual intervention. While the removal of eyeblink artifacts from multichannel EEG data has been achieved mainly through the use of ICA, several studies have demonstrated that the independent components associated with eyeblink artifacts contain leaked neuronal information. As the number of input EEG channels decreases, the leakage worsens, resulting in the loss of important EEG data. To overcome this challenge, a new framework that combines ICA with continuous wavelet transform is proposed in Maddirala and Veluvolu (2022). Compared to state-of-the-art research, the results in Maddirala and Veluvolu (2022) demonstrate that the technique achieves reliable separation for EEG signals with fewer channels while preserving all neuronal information. After the initial outlier thresholding of EEG data, a new technique was proposed in Sheoran and Saini (2020) that combined the noise-adjusted principal component transform (NAPCT) and canonical correlation analysis (CCA). The proposed automatic electrooculogram and eyeblink artifact correction of EEG signals using the CCA and NAPCT approach successfully removed artifactual components without manual intervention. The automated CCA-multichannel wiener filter (CCAMWF) technique is an automatic and unsupervised method for identifying and removing ocular artifacts, as proposed in Miao et al. (2022). The results show that the CCAMWF is a viable method for eliminating EOG artifacts from emotional EEG data. Furthermore, EEG signal processing techniques such as blind source separation and wavelet transform for multichannel EEG denoising have been proposed (Rakibul Mowla et al., 2015; Yan et al., 2022). Wavelet transform and signal denoising using the wavelet method and a comparative study of wavelet-based unsupervised ocular artifact removal techniques for single-channel EEG data were presented in Dautov and Özerdem (2018) and Khatun et al. (2016). Other EEG signal processing techniques, such as ECG signal denoising by discrete wavelet transform (Aqil et al., 2017), EEG based on wavelet transform and ICA (Zhou and Gotman, 2004), ocular artifact denoising from EEG using notch filter and wavelet transform (Tibdewal et al., 2016), EEG signal preprocessing based on empirical mode decomposition (Zhang et al., 2008), EEMD-CCA approach to removing muscle artifacts for pervasive EEG (Chen et al., 2019), and EEG signal denoising using optimal wavelet transform (Alyasseri et al., 2019) have been proposed to demonstrate the impact of EEG signal processing techniques. These studies indicate that EEG signal-processing techniques can potentially improve EEG analysis and diagnosis, leading to better emotion classification results.

Feature extraction is crucial for categorizing emotions using EEG signals. The quality of the feature extraction directly affects the accuracy of emotion categorization. The primary method used in traditional EEG emotion research is artificial EEG feature extraction related to emotions, for example, obtaining the energy ratio between various frequency bands and the power spectrum of a particular frequency band. Although this approach is straightforward, its generalization capacity diminishes when the extracted feature level is low, limiting improvements in classification accuracy and necessitating substantial experience and expertise from EEG analysts. Recent EEG emotion studies have proposed novel techniques for extracting features from EEG signals. The methods that are most frequently employed include the empirical mode decomposition, sample entropy, Differential Entropy, Wavelet Entropy, Power Spectral Density, statistical, and Short-Time Fourier Transform (STFT) (Li et al., 2017; Liu et al., 2021; Piho and Tjahjadi, 2020; Veeramallu et al., 2019; Wei et al., 2020; Zheng et al., 2021; Zhuang et al., 2017). The time and frequency domains are the primary focus of traditional EEG feature studies (Wang and Wang, 2021). Nonetheless, given the non-linear nature of EEG signals, more in-depth research can be conducted by analyzing EEG signals using non-linear dynamics. The two fundamental non-linear feature analyses are information theory and chaos theory. The Hurst index, maximal Lyapunov index, and correlation dimension are examples of chaos theory. Permutation entropy, power spectrum entropy, singular value decomposition entropy, sample entropy, and other entropies are examples of information theory techniques. Hu et al. (2021) extracted effective data-driven spectrogram-like features from the raw EEG signals. This approach addresses the limitations of previous approaches that require hand-extracted features or their approximations by utilizing convolutional kernels scaled from a data-driven pattern to disclose a frequency-like dimension. Data enrichment and feature extraction approaches were suggested by Kalashami et al. (2022) to address the high dimensionality and lack of data. The authors utilized a Conditional Wasserstein Generative Adversarial Network (GAN) to add more EEG features to the retrieved features. A new practical EEG feature called differential entropy was proposed by Duan et al. (2013) to represent the characteristics associated with emotional states. The results indicated that differential entropy is more suitable for emotion recognition compared to traditional energy-spectrum features. A multi-method fusion of cross-subject emotion recognition based on high-dimensional EEG features was presented in Yang et al. (2019). Several features were removed to create high-dimensional features. Subsequently, an

efficient technique for cross-subject emotion recognition was developed using high-dimensional characteristics. Moreover, an improved EEG-based emotion recognition through information enhancement in a connectivity feature map has been introduced (Akhand et al., 2023). The novel approach suggested by the authors clearly integrates the connectivity feature map measurements of the two separate methodologies, and the resultant fused connectivity feature map offers more relevant results. Such connectivity feature map fusion does not incur additional computational expenses when training machine learning models. Emotion recognition using three-dimensional features and convolutional neural networks (CNNs) from multichannel EEG signals has been proposed (Chao and Dong, 2021). The approach proposed by the authors first processes each channel's EEG signal in the time domain to obtain time-domain features. Subsequently, based on the electrode sensor's locations, the features of each channel were handled as a three-dimensional (3D) feature matrix. The 3D feature matrix was then processed using an advanced CNN equipped with univariate and multivariate convolution layers for emotion recognition.

Recently, machine learning algorithms have garnered significant attention for EEG-based emotion identification (Doma and Pirouz, 2020). Several well-known machine learning algorithms are currently extensively employed in the study of EEG emotion classification, including the SVM, KNN, RF, and GNB (García-Hernández et al., 2023; Masrur Ahmed and Tanzim Sabur, 2023). Without considering the inherent temporal dynamic information of EEG signals, these superficial categorization methods rely solely on the features of the EEG signals. A significant amount of pre-knowledge is required for traditional machine-learning algorithms to identify the properties of EEG signals. Nonetheless, the intricate cognitive processes of many individuals differ significantly, making it challenging to identify traits that are both effective and representative. Increasing EEG signal categorization accuracy is exceedingly difficult because of these factors. In some studies (Albraikan et al., 2019; AlZoubi et al., 2012; Asghar et al., 2020; Liu et al., 2018; Putra et al., 2018), the authors effectively used a KNN classifier for EEG emotion classification. The KNN classifier is an easy-to-use and established machine-learning technique used for EEG emotion categorization. Assigning a label to an unknown sample based on the majority class among its k nearest neighbors in the feature space is the basic idea behind KNN. This algorithm uses electrical activity captured from the brain to identify emotional states in the context of EEG emotion classification. Despite being a simple and understandable method, effectiveness of KNN depends on the characteristics of the EEG data, the quality of the feature representation, and the choice of parameters. Researchers have frequently investigated alternative machine learning methods and hybrid models to improve the precision and resilience of emotion classification using EEG signals. An SVM-based machine learning classifier was used for emotion recognition in studies by Chao and Dong (2021), Gao et al. (2020), Li et al. (2020), Liu et al. (2018), Nawaz et al. (2020), Zhao et al. (2018), and Zhang et al. (2016). In these studies, SVM classifiers successfully classified emotions from EEG data because of their capacity to manage imbalanced datasets, optimize the margins between classes, and handle high-dimensional data. However, thorough parameter adjustment and kernel selection are essential for obtaining the best results in certain EEG emotion classification tasks. Furthermore, RF-based EEG emotion recognition was proposed by combining valence lateralization and ensemble learning with tuning parameters (Pane et al., 2019). The results of this study show that valence emotion lateralization significantly improves the performance of EEG-based emotion recognition. This suggests that the left hemisphere is more predominant in producing happy and calm emotions, whereas the right hemisphere is more adept at producing angry and depressed emotions. A GNB-based machine learning classifier is presented in Dabas et al. (2018). When using a GNB-based technique, the resulting emotional states may provide valuable insights into how participants think and behave in specific situations. A stacked-ensemble method for EEG brainwave emotion detection was proposed in Jain et al. (2021). Their study aimed to identify emotional states using stacked ensembles and machine learning. The original data and the predictions of basic models were mixed, and fresh data were fed into the stacked model. The stacked model achieved a low misclassification rate because it outperformed the basic models in terms of accuracy, precision, and recall.

Deep learning methods have shown promising results in this domain, addressing challenges such as high dimensionality, noise, and nonstationary characteristics of EEG signals (Ma et al., 2024). CNNs have been widely employed in EEG-based emotion recognition due to their ability to capture spatial features across multiple EEG channels (Guo et al., 2024; Jha et al., 2024). CNNs can efficiently extract local patterns in EEG data by modeling spatial relationships among channels, which is beneficial for emotion recognition tasks where spatial distributions of brain activity are significant. For instance, Bajaj and Requena Carrión (2023) transformed EEG signals into 2D spatio-spectral feature images, enabling CNNs to process EEG data as visual patterns, which improved emotion classification accuracy. Later studies expanded this approach by introducing 3D CNNs (Su et al., 2022), which captured both spatial and temporal dimensions, enhancing the model's ability to recognize complex emotional states from EEG data. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, are commonly used to model temporal dependencies inherent to EEG signals (Das Chakladar et al., 2024). Capturing sequential patterns becomes crucial as emotions evolve. LSTM-based models have shown effectiveness in tracking these patterns, as evidenced by the work of Yaacob et al. (2024), where LSTM networks outperformed traditional machine-learning approaches in classifying emotions from EEG signals. Chakravarthi et al. (2022) combined CNNs and LSTMs, leveraging CNNs for spatial feature extraction and LSTMs for temporal sequence learning, yielding high accuracy for multiclass emotion recognition. Researchers have developed hybrid architectures that combine CNNs, RNNs, and other deep-learning components to enhance the capabilities of individual deep-learning models (Nandini et al., 2024). These hybrid approaches leverage multiple models' strengths to address the spatial and temporal complexities in EEG data. For example, Iyer et al. (2023) proposed a CNN-LSTM hybrid model that achieved high performance by combining CNN spatial feature extraction with LSTM temporal sequence learning. Another notable approach is combined with CNNs and Graph Neural Networks (GNNS) to model spatial relationships more flexibly, thus enabling the representation of EEG channels as nodes in a graph structure (Gao et al., 2024). This approach captures complex interactions between EEG channels, which are essential for understanding emotional states.

Initially designed for natural language processing, transformer models have recently been adapted for EEG-based emotion recognition owing to their strong ability to capture long-term dependencies and complex relationships across data (Guo et al., 2022; Wei et al., 2023). Unlike RNNs, transformers use self-attention mechanisms to model temporal dependencies in parallel, improving computational efficiency and potentially improving classification accuracy. Wang et al. (2022) applied transformer models to EEG data by encoding temporal dependencies and reported superior performance compared with traditional deep learning models. Transformer models often require large datasets and computational resources, which can be challenging for EEG-based applications. Attention mechanisms have been incorporated into deep-learning models to focus on the most relevant segments of EEG data for emotion recognition (Gunda et al., 2024). Attention mechanisms dynamically allocate weights to different parts of the input sequence, enabling the model to focus on specific time points or channels that contribute the most to the emotional states (Gong et al., 2023). For example, Liu et al. (2022) introduced a spatiotemporal transformer model that applied attention to both spatial and temporal dimensions, enhancing the interpretability and accuracy of the model in emotion classification tasks. These attention-based models are particularly effective in dealing with the high variability and noise in EEG signals because they prioritize meaningful signal patterns.

The scarcity of labeled EEG data has led researchers to explore generative models such as GANs to augment training datasets (Habashi

et al., 2023). GANs generate synthetic EEG samples that resemble real data, thereby enhancing the generalization ability of the model in low-data regimes (Qiao et al., 2024). This approach improves the training efficiency and addresses the data imbalance that often arises in multiclass emotion recognition tasks (Bao et al., 2021; Luo et al., 2020). Additionally, GANs and other generative models can assist in transfer learning applications, where synthetic EEG samples help bridge domain gaps between EEG datasets or emotion recognition tasks.

In summary, research on EEG-based emotion recognition has progressed from basic experiments that prove the theory to complex studies that leverage machine learning models, advanced signal processing, deep learning models, hybrid models, transformers, attention mechanisms, and practical applications. In this active field of research, issues still need to be explored, and innovative solutions must be found concerning generalization, model robustness, interpretability, computational efficiency, real-time processing, and ethical concerns.

3. System overview of EEG-based emotion recognition

Emotions significantly affect decision-making, social connections, and general well-being in human communication. Comprehending and deciphering emotions has been an enduring endeavor across multiple fields of study because of their intricate and diverse nature. New methods for recording and interpreting emotional states have recently emerged owing to the convergence of neuroscience and technology. One possible approach to objectively recognize emotions is the use of EEG signals. EEG signals provide real-time access to the brain's electrical activity and unique insights into the neurological correlates of emotions. Using this non-invasive neuroimaging method, one can directly observe the dynamic interaction of brain circuits linked to emotional events by measuring the electrical potentials produced by synchronized neuronal firing. EEG-based emotion identification research is expanding owing to the theory that distinct patterns of brain activity are correlated with different emotional states. The shortcomings of conventional techniques, including self-reporting and facial expression analysis, drive EEG-based emotion recognition. However, these techniques may not fully capture emotional experiences and are prone to subjective bias. However, EEG provides a direct physiological metric that does not require explicit behavioral responses. By deciphering emotional neural signatures, EEG-based methodologies can enhance the comprehension of affective processes, facilitate precise evaluations of affective states, and pave the way for novel applications in domains spanning mental health diagnostics to human-computer interactions. Fig. 1 presents a systematic overview of EEG-based emotion recognition.

An EEG-based emotion identification system effectively interprets and reacts to human emotions through an interdisciplinary approach that combines machine learning, neural networks, signal processing, and human-computer interaction. The overall workflow (Fig. 1) involves inducing emotions using visual stimuli, during which EEG signals are captured, further processing these signals, and extracting and selecting valuable features. The collection of brainwave data is the initial stage of the EEG-based emotion identification process. Electrodes were positioned on the scalp in a specific manner to record the electrical impulses produced by the brain. The location and number of electrodes vary depending on the EEG system utilized, and a higher spatial resolution is achieved with more electrodes. Various artifacts such as muscle contractions, eye movements, and outside interference frequently taint raw EEG data. Preprocessing methods are used to improve the signal quality and clean the data. This may entail filtering, artifact removal, and normalization to ensure that the ensuing analysis concentrates on the pertinent brainwave patterns. The system then performs feature extraction. Feature extraction is a vital step in the extraction of relevant information from preprocessed EEG data. Features in the time and frequency domains, such as the signal entropy, band power, and power spectral density, are frequently extracted. The system takes EEG signals across time and extracts statistical parameters (mean, variance, skewness, and kurtosis) to create time-domain characteristics. The analysis of signal frequency components is performed via frequency domain features using methods such as the Fast Fourier Transform (FFT). Time-frequency features record time-varying frequency information using wavelet transform and STFT techniques. During the feature selection phase, the system uses dimensionality reduction strategies, which lower the number of features and enhance the model performance using techniques such as principal component analysis (PCA), t-distributed Stochastic Neighbor

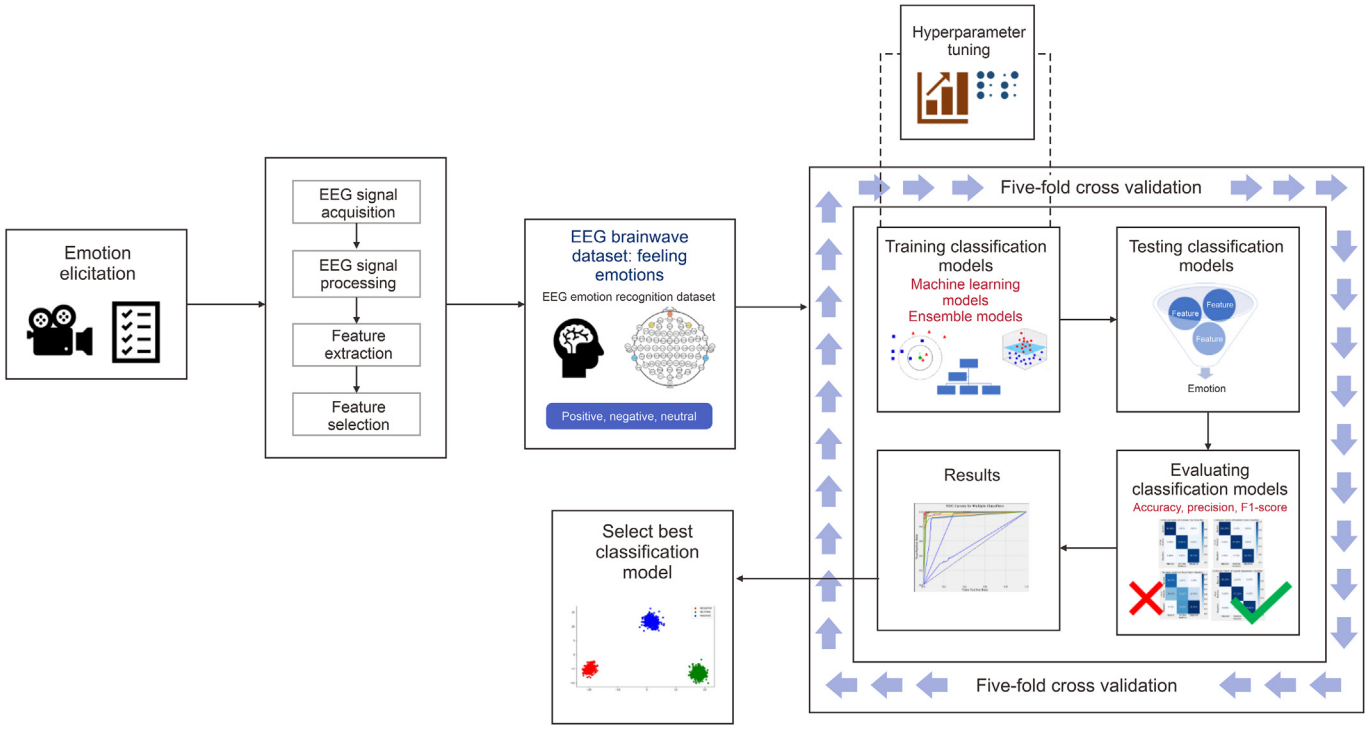

Fig. 1. System overview of electroencephalogram (EEG)-based emotion recognition.

Embedding (*t*-SNE), and Linear Discriminant Analysis (LDA) or feature selection algorithms. In the last phase, the system classifies emotions using machine learning models. Machine-learning algorithms, particularly classification models, play a central role in EEG-based emotion recognition. Supervised learning techniques, including LR, SVM, GNB, KNN, DT, and ensemble models, such as RF, AdaBoost, LightGBM, XGBoost, and CatBoost, are commonly employed to train models on lagged datasets. This study aimed to develop a robust and accurate model for classifying EEG features into specific emotional states. Background research for our comparative examination of different machine learning models for EEG-based emotion recognition is presented in the following subsections. By focusing on optimizing the model performance and interpretability, our work addresses critical gaps in the existing literature regarding the application of machine learning techniques to EEG datasets.

3.1. EEG brainwave dataset

EEG brainwave dataset: The feeling emotions dataset (Bird et al., 2019) was used in this study to classify discrete emotions. In this dataset, the participants viewed video clips, as shown in Table 1 (5 short and 5 longer, typically between 1.5 and 4 s), to simulate three emotions: negative, neutral, and positive. Six minutes for each state (positive, neutral, and negative) were recorded from two adults (highly educated STEM students), 1 male and 1 female aged 21 ± 1, producing 36 min of brainwave activity data. The dataset was fully anonymized and publicly accessible. No new data involving human participants was collected or processed in this study.

The Muse headset was used to gather data from the extracranial electrodes at TP9, AF7, AF8, and TP10. Thus, human emotions were extracted using visual stimuli, and data were recorded from the EEG sensors based on 10–20 electrode placements. The EEG electrode sensor-based setup is shown in Fig. 2. This study classified all three emotions. These emotions are positive, negative, and neutral, as shown in Fig. 3. The emotion graph indicates that the signal patterns for each emotion are different, and the difference in signal characteristics can be used as the key component of emotion classification. The data distribution shown in Fig. 4 indicates that the EEG data samples were equally balanced and that our system did not face any class imbalance problems during model training and testing. When the dataset is balanced, each target class is represented by the same number of input samples.

Using this publicly available dataset, we highlighted the unique attributes relevant to our study's goals. Although commonly used, this dataset offers distinct advantages for analyzing EEG-based emotional states and capturing positive, neutral, and negative emotions under consistent experimental conditions. The recordings, taken from multiple electrode sites using the Muse headband, provide a detailed set of temporal statistical features, such as mean and variance, along with frequency-domain features obtained through the FFT. These features enable a nuanced view of brainwave patterns, facilitating a comprehensive exploration of machine-learning methods for emotion classification. The dataset design aligns well with our focus on emotion classification, allowing us to rigorously evaluate and generalize machine learning techniques in EEG-based emotion recognition.

Table 1
Stimuli used to elicit emotions.

Movie clip	Film production	Description	Emotion elicited
Marley and Me	Twentieth Century Fox	Death scene	Negative
Up	Walt Disney Pictures	Opening death scene	Negative
My Girl	Imagine Entertainment	Funeral scene	Negative
La La Land	Summit Entertainment	Opening musical number	Positive
Slow Life	BioQuest Studios	Nature timelapse	Positive
Funny Dogs	MashupZone	Funny dog clips	Positive

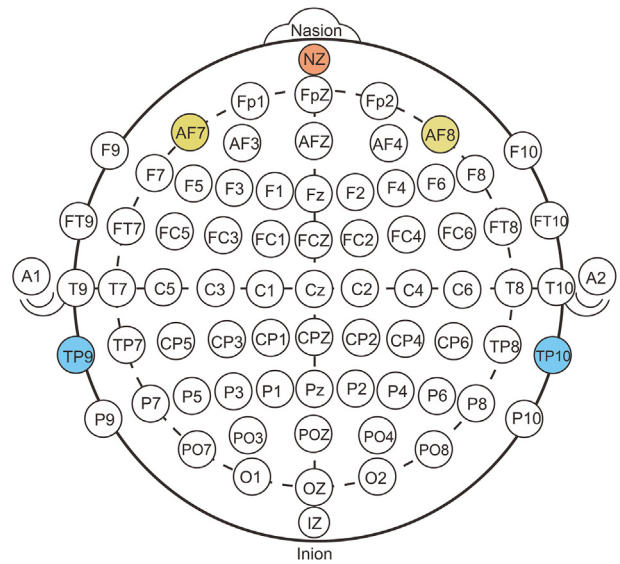

Fig. 2. Electroencephalogram (EEG) electrode sensor TP9, TP10, AF7, AF8.

3.2. Dimensionality reduction techniques

We explore various dimensionality reduction techniques in the context of time-series data, such as PCA, *t*-SNE, and LDA. PCA is a linear dimensionality-reduction technique that identifies the principal components of a dataset, which are linear combinations of the original features. It projects data onto a lower-dimensional subspace while maximizing the variance of the data. The PCA works well when the underlying structure of the time-series data is linear. This can reduce the dimensionality of the multivariate time-series data and remove noise and redundancy. However, it may not capture non-linear patterns in the data. *t*-SNE is a non-linear dimensionality reduction technique that focuses primarily on visualizing high-dimensional data in a lower-dimensional space. It attempts to maintain the pairwise similarities in data points in the original space. *t*-SNE is effective at preserving local structures in data and visualizing clusters or patterns within time-series data. LDA is a supervised dimensionality-reduction technique that aims to determine linear combinations of features that best separate different classes in a classification problem. It identifies the dimensions that maximize between-class variance (separability) while minimizing within-class variance. LDA can be useful for labeling time-series data and reducing dimensionality while maintaining class separation.

3.3. Emotion classification

Several machine-learning models have been implemented to classify EEG signals for human emotion recognition. These methods include LR, SVM, GNB, DT, and KNN, which are widely used as classification methods for EEG-based emotion recognition systems. In ensemble model classification, the system extracts features with different transformations and combines two or more learning algorithms to improve the predictions. It then combines the predictions of the enhanced performance through adaptive voting schemes. RF, AdaBoost, LightGBM, XGBoost, and CatBoost are ensemble models implemented to classify EEG signals for human emotion recognition. The following paragraphs briefly explain each machine learning model and ensemble classifiers used in our comparative analysis.

3.3.1. LR

LR is a supervised learning method used for classification. It models the probability of a given input belonging to a particular class. The prediction was based on a joint probability function. Assuming that all the data points are independent and identically distributed, the joint

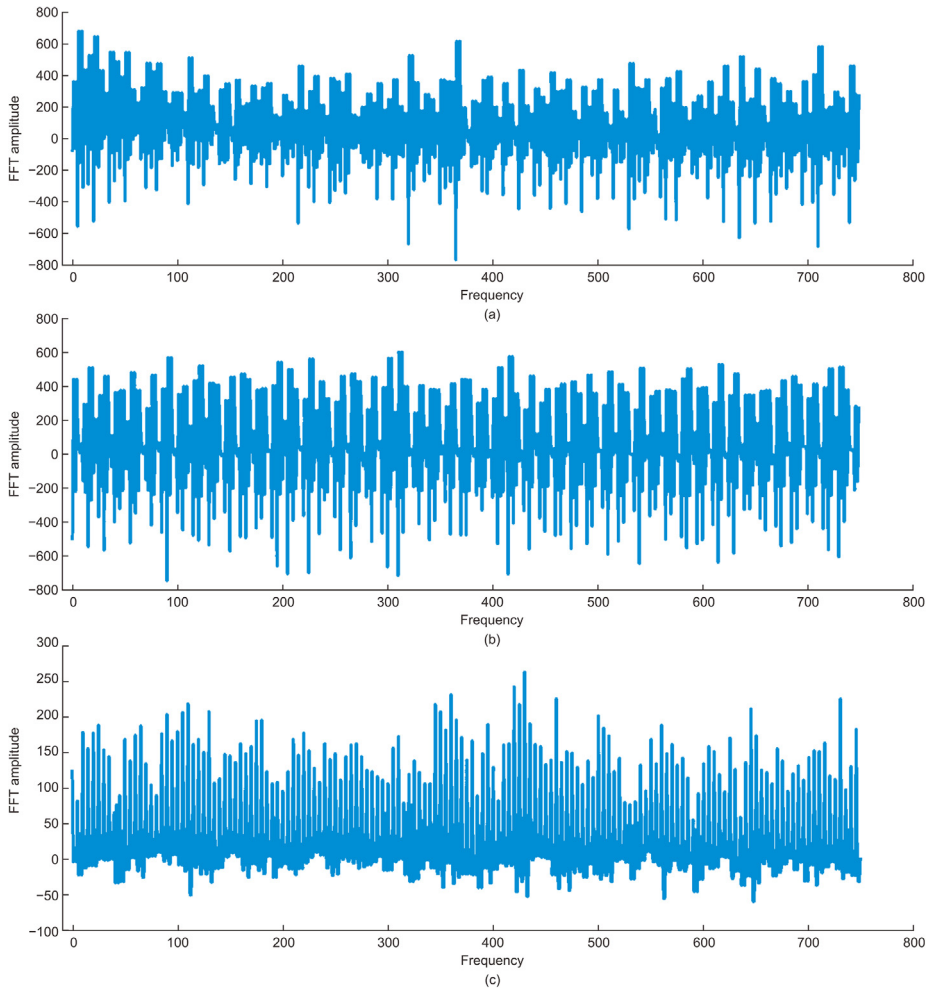

Fig. 3. Electroencephalogram (EEG) brainwave dataset emotions. (a) Positive; (b) negative; (c) neutral. FFT: fast Fourier transform.

probability is used for correct classification and to provide a single output. It maps the input features to a value between zero and one using a logistic function (also known as the sigmoid function) as follows:

$$\text{Logistic function} = \frac{1}{1 + e^{-k(x - x_0)}} \tag{1}$$

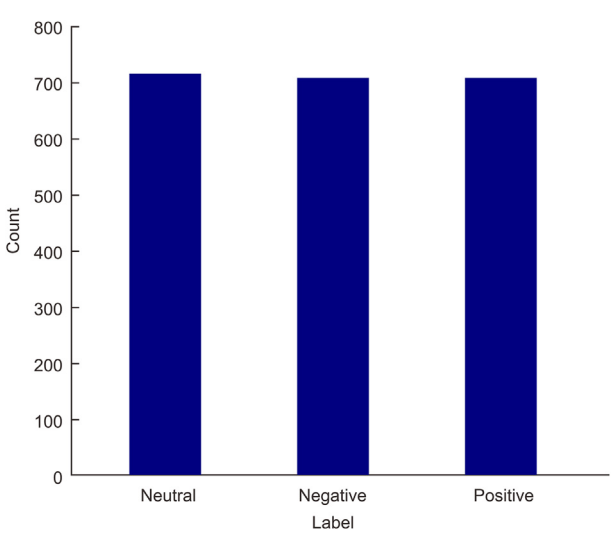

Fig. 4. The bar plot shows balanced data distribution. When the dataset is balanced, each target class is represented by the same number of input samples.

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-x}} \tag{2}$$

For EEG-based emotion recognition, LR is a valuable method that compromises the ease of use and potency. Owing to their interpretability, researchers and experts can learn more about how particular EEG characteristics contribute to the identification of various emotional states. Even with more sophisticated models, LR remains a sensible option, especially when emotion-recognition applications demand a transparent and understandable model.

3.3.2. SVM

The SVM is a supervised learning method for solving linear and nonlinear regression and classification problems. The primary objective was to identify hyperplanes that accurately separate distinct groups within n-dimensional data comprising two or more classes. By considering the designated labels or classes, SVM attempts to optimize the separation boundaries among data points. These boundaries led to the identification of an optimal hyperplane characterized by the maximum distance to the nearest training point of any class, ensuring effective feature separation. When linear separation of data is unattainable through an SVM, kernel functions are employed to transform the input data into high-dimensional feature spaces. These transformations allow a hyperplane to facilitate linear separation in a high-dimensional space. Kernel functions such as polynomial, linear, and Gaussian functions play pivotal roles in SVM. The effectiveness of the SVM-based classification depends on parameter optimization and the informed choice of a suitable kernel function. Incorrect SVM parameter settings can lead to suboptimal

classification outcomes, manifesting as over-fitting or under-fitting. SVM offers the advantage of managing overfitting by adjusting the data separation margin represented by support vectors. SVM is well known for classifying EEG brainwave datasets because of its ability to handle high-dimensional data and find complex decision boundaries. However, the effectiveness of SVMs depends on various factors, including the choice of kernel, feature representation, and nature of the EEG data. Experimentation and thorough analysis are essential to create an effective SVM model for EEG-based emotion recognition.

3.3.3. GNB

The GNB classifier is a classification algorithm based on Bayes' theorem. It follows from the theorem that the occurrence of an instance is independent of other cases; that is, the features are classified independently of one another. GNB assumes that each class follows a Gaussian distribution, also called a normal distribution, which is defined as:

$$P(A|B) = \frac{1}{\sqrt{2\pi\sigma_B^2}}\exp\left(-\frac{(A-\mu B)^2}{2\sigma_B^2}\right) \tag{3}$$

The μ is defined as the mean, and σ is the standard deviation. GNB is particularly useful when dealing with relatively small datasets and when the independence assumption holds reasonably well. While it may not capture complex dependencies between features, it can serve as a fast and interpretable baseline model for EEG-based emotion recognition tasks. Experimentation and careful evaluation are essential for determining its suitability for an EEG dataset and emotion recognition.

3.3.4. DT

The DT classifier illustrates decisions using a flowchart-style structure by recursively dividing the data based on feature values. This culminates in a prediction or decision made at the leaf nodes. This helps interpret and explain the model decisions. This classifier is a potent tool for classification tasks, including EEG-based emotion recognition. The classifier is adaptable and can capture intricate decision boundaries. However, the classifier is prone to overfitting, and hyperparameter tuning is necessary to optimize performance. Experimentation, evaluation, and possibly ensemble methods (e.g., RFs) are worth considering to enhance the robustness and generalization capabilities of the model in EEG-based emotion recognition tasks.

3.3.5. KNN

KNN is a supervised machine-learning algorithm that categorizes data samples by considering their proximity, operating under the assumption that closeness implies similarity. This is a nonparametric instance-based classification algorithm that can be used for EEG-based emotion recognition. Although the KNN is an easy-to-understand algorithm, its performance can be affected by the feature space, number of neighbors, and distance metrics selected. It usually performs well when the dataset is not too large, and the decision boundaries are reasonably smooth. Nevertheless, they may be susceptible to distractions or loudness. Similar to any other classifier, experimentation and a thorough review are essential to determine an algorithm's eligibility for a particular EEG-based emotion recognition task.

3.3.6. RF

RF is an ensemble that combines the predictions from multiple individual DTs to obtain more accurate predictions. Each tree classifies the data into target classes based on the attributes, and the class into which it is classified is the final prediction. Compared to individual DTs, RFs are recognized for their resilience, ability to manage nonlinearity, and ability to reduce overfitting. They are standard options for classification tasks and are suited to various data formats, including EEG signals. Trial and error, assessment, and potential hyperparameter adjustments are required to obtain an optimal RF model for EEG-based emotion identification.

3.3.7. AdaBoost

Adaptive Boosting is an algorithm that creates multiple unique instances of one type of model prediction to improve the model performance in situations where the selected parameters are less effective (Schapire, 2013). AdaBoost sequentially trains a series of weak learners with increasing emphasis on misclassified data points from previous learners and then combines predictions through weighted voting to improve on a prior iteration of the model. The model combines the predictions from multiple weak learners (typically DTs) to create a robust model. This model is known to improve the accuracy of weak learners, making it particularly useful for combining multiple simple models into robust and accurate models. It can adapt to complex decision boundaries and is less prone to overfitting. Optimizing the AdaBoost model for EEG-based emotion identification requires testing, assessing, and adjusting the hyperparameters.

3.3.8. LightGBM

Boosting combines trees sequentially using the previous tree's information. Boosting forms the basis of gradient boosting, which minimizes the loss function. LightGBM (Ke et al., 2017) is a gradient-boosting framework that is designed to be fast and efficient, particularly for large datasets. The LightGBM uses a histogram-based approach to build trees that can directly handle categorical features. Its speed and capacity to handle categorical variables make it appropriate for machine-learning applications, such as identifying emotions using EEG signals. The classifier performs testing, assessment, and potential hyperparameter adjustments to get the best LightGBM model for EEG-based emotion identification.

3.3.9. XGBoost

XGBoost (Chen and Guestrin, 2016) is a highly efficient implementation of gradient boosting. The model includes techniques like regularization and parallel processing, making it faster and more accurate. XGBoost works well on a wide range of machine-learning tasks and is versatile. It is a potent option for EEG-based emotion detection because of its capacity to accommodate missing values, incorporate regularization, and produce findings that can be understood. It is crucial to experiment with, evaluate, and adjust the hyperparameters when optimizing the XGBoost model for EEG-based emotion recognition.

3.3.10. CatBoost

CatBoost (Prokhorenkova et al., 2018) is a gradient-boosting library designed to effectively handle categorical features. Instead of converting categories into numerical values using one-hot or label encoding, CatBoost directly processes categorical features using ordered target statistic encoding. Compared with other gradient boosting methods, CatBoost is intended to handle categorical features effectively and frequently requires less hyperparameter tuning. It is crucial to conduct experimentation, evaluation, and hyperparameter tuning to optimize the CatBoost model for EEG-based emotion recognition.

3.4. Feature selection

A crucial step in creating EEG-based emotion recognition systems is feature selection, which isolates the most pertinent and instructive elements of EEG signals for precise emotion categorization. Effective feature selection can increase interpretability, reduce computing complexity, and improve model performance in EEG data that frequently contain channels and temporal points. Notably, the efficacy of feature-selection techniques may differ depending on the particular attributes of the

EEG dataset and the underlying emotional states being identified. We must experiment and validate our dataset to find the best feature selection strategy for an EEG emotion detection task. The standard methods used in feature selection include statistical tests (*t*-test and ANOVA), filter methods (CF and information gain), wrapper methods (RF and genetic algorithms), and embedded methods (LASSO and tree-based methods). Among these feature selection methods, we implemented LASSO feature selection, which analyzes the correlation matrix of selected features to provide insights into their relationships. The LASSO feature selection method is a regularization technique that adds a penalty term to the linear regression objective function, forcing some coefficients to be exactly zero. This results in a sparse model with fewer features, yielding sparse feature selection in which only a subset of features with nonzero coefficients is retained. The objective function for LASSO includes a regularization term that penalizes the absolute values of the coefficients, defined as:

$$minimize J(\beta) = \frac{1}{2n}\|Y - X\beta\|_2^2 + \lambda\|\beta\|_1 \tag{4}$$

Here, Y is the target variable (emotion labels), X is the feature matrix, β is the vector of coefficients, $\|\|_2$ is the L2 norm, and $\|\|_1$ is the L1 norm. The hyperparameter λ controls the strength of the regularization. Implementing LASSO feature selection in EEG-based emotion recognition allows automatic identification and prioritizes the most informative features, improving model efficiency and interpretability.

3.5. Hyperparameter tuning

Hyperparameter tuning is a critical first step in maximizing the effectiveness of EEG-based emotion identification systems. Hyperparameters are the external model configuration settings for machine learning that are set before training and are not learned from the data. Finding the ideal set of hyperparameter values is the first step in improving the capacity of the model to generalize to fresh, untested inputs. Usually, there is a tradeoff between model performance and execution time because the optimal hyperparameters may require more iterations or computational resources to converge. The steps used in hyperparameter tuning for an EEG-based emotion recognition system are as follows:

(1) Define hyperparameters: Determine which hyperparameters must be adjusted for a particular emotion recognition model. This could include regularization, learning rates, and model architecture parameters.
(2) Split the dataset: Make training, validation, and testing sets. The model was trained on the training set, the hyperparameters were adjusted on the validation set, and the model's ultimate performance was assessed on the testing set.
(3) Select a performance metric: A suitable performance measure is selected to assess the model while fine-tuning its hyperparameters. When assessing emotion recognition, metrics such as accuracy, precision, recall, and F1-score are frequently employed.
(4) Choose a search space: Establish the range of values to be investigated for each hyperparameter during tuning. This can refer to a certain range of regularization strengths, learning rates, or other variables.
(5) Hyperparameter tuning methods: Hyperparameter tuning methods include grid search, random search, and Bayesian optimization. We can use one or a combination of hyperparameter tuning methods to search the hyperparameter space. The system manually specifies a hyperparameter grid in the grid search method and checks the model performance. In a random search, hyperparameter values are randomly sampled from predefined distributions, and the Bayesian optimization method models the performance metric as a probabilistic surrogate and iteratively selects parameters to optimize the surrogate.
(6) Evaluate on validation set: The model's performance was evaluated using the chosen performance metric on the validation set for each set of hyperparameters.
(7) Final evaluation on test set: Once the parameters were adjusted, the final model's generalization performance was tested on the test set.
(8) Iterative refinement: If refinement of the parameter values is required based on the validation set performance, repeat steps 5–7.

By systematically tuning the hyperparameters, we enhanced the generalization performance of our EEG-based emotion recognition model, thereby ensuring its robustness and effectiveness across diverse datasets.

3.6. k-fold cross-validation

Cross-validation is a statistical method used to assess the performance of a trained model using unseen data. Cross-validation involves splitting the dataset into k subsets (folds), training the model on k-1 folds, and evaluating it on the remaining fold. This process was repeated k times, with each fold as the test set. By evaluating each model's performance in several ways, the approach assists in identifying the optimal model. When used in an EEG-based emotion identification system, k-fold cross-validation helps ensure that the particular way the dataset is divided into training and testing sets did not significantly affect the performance of the model. This technique offers a systematic method for evaluating the generalization of the model on various subsets of the EEG dataset. It provides a more thorough assessment of the model's performance by considering various data splits. The findings prove that a specific training-testing split is unnecessary for the model to function well. Creation and assessment of EEG-based methods, the k-fold cross-validation significantly improves the systems' overall performance, robustness, and dependability.

4. Experiment results and analysis

This section presents the experimental results and comprehensive analysis. Our analysis plays a pivotal role in assessing the efficacy and reliability of EEG-based emotion-recognition systems. The experimental outcomes validated the performance of the model and provided insights into the interpretability of the results. EEG-based emotion recognition is a multifaceted process that involves assessing both quantitative and qualitative aspects of model behavior. A thorough examination of the results enhances the credibility of this study and provides a foundation for further advancements in emotion recognition using EEG data. Our experimental results and analyses include performance metrics, cross-validation results, confusion matrix, hyperparameter tuning, interpretability, visualization, limitations, and future research directions. The following subsections present a comparative analysis of various supervised machine-learning algorithms.

4.1. Performance metrics

The computation of multiple performance metrics addresses EEG-based emotion-recognition models. The standard performance metrics included F1-score, area under the receiver operating characteristic (AUC-ROC) curve, recall, accuracy, and precision. These metrics quantify the model's ability to identify different emotional states correctly. Each class represents a distinct category in a multiclass EEG-based emotion-classification scenario. True positives (TP) occur when the model correctly identifies instances of a specific class and true negatives (TN) occur when

it accurately recognizes instances not belonging to that class. False positives (FP) arise from misclassifying instances as belonging to a class, whereas false negatives (FN) occur when instances of the class are mistakenly predicted as not belonging to it. Performance matrices such as the accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC curve are defined as follows:

4.1.1. Accuracy

Accuracy measures how well the model predicts the label or class of each input. In multiclass classification, the accuracy is the ratio of correctly predicted instances to the total instances, which is defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{FP}}{\text{TN} + \text{TP} + \text{FP} + \text{FN}} \tag{5}$$

4.1.2. Precision (P)

Precision assesses the accuracy of positive predictions made by the model which is given as:

$$\text{P} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

4.1.3. Recall (R)

Recall measures the ability of the model to capture instances of a specific class.

$$\text{R} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

4.1.4. F1-score

F1-score is the harmonic mean of precision and recall and offers a balanced measure of a model's performance across all classes.

$$\text{F1-score} = 2 \times \frac{\text{P} \times \text{R}}{\text{P} + \text{R}} \tag{8}$$

4.1.5. Confusion matrix (CM)

The CM summarizes the true-positive, true-negative, false-positive, and false-negative counts for each class, aiding in the detailed evaluation of the model's performance across multiple categories.

4.1.6. AUC-ROC curve

The AUC-ROC curve provides an overall performance measure by considering the ability of the model to discriminate between different classes. It assesses the trade-off between true-positive and false-positive rates across multiple class comparisons. A higher AUC-ROC score (close to 1) indicates a better overall performance in distinguishing between various classes.

4.2. Hyperparameter tuning

If hyperparameter tuning is performed during the experiment, the results should include the optimal hyperparameter values and their impact on the model's performance. This analysis helps understand the model's sensitivity to different hyperparameter choices. We used GridSearch to perform hyperparameter tuning during the experiment, and the best hyperparameters were selected from the hyperparameter grid, as listed in Table 2.

Table 2 lists the best hyperparameters through a grid search for various supervised machine-learning models. Each row corresponds to a specific model, and the columns represent a hyperparameter grid and its corresponding optimized value. The results, including the optimal hyperparameter values and their impact on the model performance, are presented in Tables 3 and 4.

It should be noted that the accuracies of KNN and AdaBoost increased upon tuning the hyperparameters. Simultaneously, the rest remained approximately the same, whereas the computational time was optimized for the models. The computational overhead was minimized by optimizing the hyperparameters, ensuring streamlined model training and inference pipelines across all architectures. The notable improvement in the KNN and AdaBoost accuracy post-tuning suggests that these algorithms are more sensitive to hyperparameter changes than the other models. This heightened sensitivity could be attributed to the intrinsic characteristics of these algorithms, such as KNN's dependence on the choice of the number of neighbors and distance metrics and AdaBoost's dependence on parameters such as learning rate and the number of weak learners. The stability of the performance of the other models, including LightGBM, XGBoost, CatBoost, RF, LR, SVM, DT, and GNB, implies that their default or manually selected hyperparameters are already well

Table 2
Summary the best hyperparameter values found while tuning supervised machine-learning models.

Model	Best hyperparameters	Accuracy (%)	Hyperparameter grid
LightGBM	'learning_rate': 0.1 'max_depth': 5 'n_estimators': 100, 'num_leaves': 31	99	'n_estimators': [50, 100] 'learning_rate': [0.01, 0.1] 'max_depth': [5, 10] 'num_leaves': [31, 63]
XGBoost	'learning_rate': 0.1 'max_depth': 5 'subsample': 0.5	99	'max_depth': [3, 5] 'learning_rate': [0.01, 0.1] 'subsample': [0.5, 0.7]
CatBoost	'depth': 6 'iterations': 100 'learning_rate': 0.1	99	'iterations': [50, 100] 'learning_rate': [0.01, 0.1] 'depth': [4, 6]
Random forest (RF)	'max_depth': None 'min_samples_leaf': 1 'min_samples_split': 2 'n_estimators': 100	98	'n_estimators': [50, 100, 200] 'max_depth': [None, 10, 20] 'min_samples_split': [2, 5, 10] 'min_samples_leaf': [1, 2, 4]
Logistic regression (LR)	'C': 1, 'max_iter': 300 'penalty': 'l1' 'solver': 'liblinear'	97	'C': [0.001, 0.01, 0.1, 1, 10, 75] 'penalty': ['l1'] 'solver': ['liblinear'] 'max_iter': [100, 200, 300, 500]
k-nearest neighbor (KNN)	'n_neighbors': 1 'p': 1, 'weights': 'uniform'	97	'n_neighbors': [1, 3, 5, 7, 9] 'weights': ['uniform', 'distance'] 'p': [1, 2]
Decision trees (DT)	'criterion': 'entropy' 'max_depth': None 'max_features': None 'min_samples_leaf': 2 'min_samples_split': 5	96	'criterion': ['gini', 'entropy'] 'max_depth': [None, 10, 20, 30] 'min_samples_split': [2, 5, 10] 'min_samples_leaf': [1, 2, 4] 'max_features': ['auto', 'sqrt', 'log2', None]
Support vector machine (SVM)	'C': 10 'gamma': 'auto' 'kernel': 'rbf'	96	'C': [0.1, 1, 10] 'kernel': ['linear', 'rbf', 'poly'] 'gamma': ['scale', 'auto'] + [0.001, 0.01, 0.1, 1]
AdaBoost	'learning_rate': 0.01 'n_estimators': 50	92	'n_estimators': [50, 100, 200] 'learning_rate': [0.01, 0.1, 1.0]
Gaussian Naive Bayes (GNB)	'priors': None 'var_smoothing': 1e-05	65	'priors': [None, [0.2, 0.3, 0.5], [0.3, 0.4, 0.3], [0.1, 0.1, 0.8]] 'var_smoothing': [1e-9, 1e-8, 1e-7, 1e-6, 1e-5, 1e-4, 1e-3, 1e-2, 1e-1]

Table 3
Classification results before hyperparameter tuning.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time for training (s)
XGBoost	99.06	99.08	99.06	99.06	33.82
CatBoost	99.06	99.08	99.06	99.06	1,681.48
LightGBM	98.59	98.62	98.59	98.59	138.61
Random forest (RF)	98.36	98.39	98.36	98.36	4.59
Logistic regression (LR)	96.72	96.75	96.72	96.70	4.81
Support vector machine (SVM)	94.15	94.22	94.15	94.09	1.97
Decision tree (DT)	94.61	94.63	94.61	94.62	3.94
k-nearest neighbor (KNN)	93.44	93.43	93.44	93.39	0.01
AdaBoost	66.28	52.34	66.28	56.44	32.72
Gaussian Naive Bayes (GNB)	65.11	63.80	65.11	62.45	0.07

Table 4
Classification results after hyperparameter tuning.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time for training (s)
LightGBM	98.59	98.62	98.59	98.59	64.47
XGBoost	98.59	98.62	98.59	98.59	57.95
CatBoost	98.13	98.15	98.13	98.12	172.26
Random forest (RF)	98.13	98.16	98.13	98.13	4.43
Logistic regression (LR)	97.19	97.23	97.19	97.19	2.43
Support vector machine (SVM)	96.25	96.25	96.25	96.23	1.97
Decision tree (DT)	96.96	96.98	96.96	96.95	5.68
k-nearest neighbor (KNN)	97.42	97.44	97.42	97.42	0.01
AdaBoost	91.57	92.43	91.57	91.51	31.21
Gaussian Naive Bayes (GNB)	65.34	64.02	65.34	62.66	0.18

Table 5
Classification performances of various hyperparameter-tuned supervised machine-learning algorithms for an 80:20 train-test split dataset (one-fold analysis).

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time for training (s)
LightGBM	98.59	98.62	98.59	98.59	64.47
XGBoost	98.59	98.62	98.59	98.59	57.95
CatBoost	98.13	98.15	98.13	98.12	172.26
Random forest (RF)	98.13	98.16	98.13	98.13	4.43
Logistic regression (LR)	97.19	97.23	97.19	97.19	2.43
Support vector machine (SVM)	96.25	96.25	96.25	96.23	1.97
Decision tree (DT)	96.96	96.98	96.96	96.95	5.68
k-nearest neighbor (KNN)	97.42	97.44	97.42	97.42	0.01
AdaBoost	91.57	92.43	91.57	91.51	31.21
Gaussian Naive Bayes (GNB)	65.34	64.02	65.34	62.66	0.18

suited for the given dataset. These algorithms may have been initialized with configurations closer to the optimal configuration, leaving less room for significant improvement through hyperparameter tuning compared with KNN and AdaBoost. Despite the diverse nature of the algorithms used, optimizing the computational time for the models indicates successful parameter-tuning efforts to minimize the computational overhead. This optimization is essential to ensure efficient model training and inference, particularly in scenarios with large datasets or real-time processing requirements.

4.3. Cross-validation results

The k-fold cross-validation results are essential for determining how well the model generalizes to other EEG dataset subsets. Examining different fold performances helps to determine possible sources of variability and sheds light on the model's resilience. Table 5 compares the performances of the models for a single-fold (80:20) splitting test-train dataset. Table 6 summarizes the cross-validation scores and classification performances of various supervised machine learning algorithms from folds 1 to 5.

In general, ensemble models are shown to have more accuracy and training time compared with other machine learning models. LightGBM and XGBoost achieved an accuracy of 98.59%. After five-fold cross-validation, the scores improved, suggesting that the models generalized well to unseen data. When the scores obtained from each iteration of the five-fold cross-validation showed improvement, the model consistently performed better across the different data subsets. This indicated that the model learned meaningful classification patterns. The k-fold cross-validation approach eliminated any bias in splitting the training data randomly on an 80:20 basis. Empirical evidence shows that, compared with their single-model counterparts, ensemble methods achieve both higher predictive accuracy and greater stability. Specifically, LightGBM (99.30%), XGBoost (98.89%), CatBoost (98.77%), and RF (98.59%) outperform LR (97.61%), SVM (97.56%), DT (97.19%), and KNN (96.62%); AdaBoost is an exception at 92.26%, while GNB is the least accurate at 66.18%.

Tables 5 and 6 illustrate that implementing cross-validation incurs a computational cost manifested as increased computational time. This additional time arose from the repeated training and evaluation of the model across multiple folds of the dataset. Each fold iteration requires model training, evaluation, and result aggregation, thereby contributing to the overall computational overhead. The observed trade-off between performance improvement and computational time underscores the importance of balancing model efficacy with practical considerations, such as time and resource constraints. Although cross-validation enhances the reliability of a model's predictions, the associated increase in computational time may be prohibitive in scenarios where rapid model deployment or real-time processing is imperative.

Table 6
Cross-validation scores and classification performances of various hyperparameter-tuned supervised machine-learning algorithms for folds 1 to 5.

Model	Cross-validation scores (%)					Average accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
	Fold-1	Fold-2	Fold-3	Fold-4	Fold-5					
LightGBM	99.77	98.59	99.30	99.30	99.53	99.30	99.30	99.30	99.30	494.64
XGBoost	99.71	98.53	98.24	98.83	99.12	98.89	98.89	98.89	98.89	497.29
CatBoost	99.12	97.95	98.83	99.12	98.83	98.77	98.77	98.77	98.77	1,673.95
Random forest (RF)	99.12	98.24	97.95	98.24	99.41	98.59	98.42	98.42	98.42	35.59
Logistic regression (LR)	98.36	96.49	96.48	98.59	98.12	97.61	97.76	97.75	97.75	27.20
Support vector machine (SVM)	98.13	97.42	96.48	96.95	98.83	97.56	97.58	97.56	97.57	33.36
Decision tree (DT)	96.96	96.49	98.12	97.65	96.71	97.19	97.37	97.37	97.37	32.42
k-nearest neighbor (KNN)	96.49	95.32	96.95	96.24	98.12	96.62	96.68	96.62	96.60	26.83
AdaBoost	92.67	92.08	90.62	91.79	94.13	92.26	92.85	92.26	92.21	234.87
Gaussian Naive Bayes (GNB)	66.04	68.15	63.85	67.61	65.26	66.18	66.16	66.18	64.39	1.77

4.4. Confusion matrix

A thorough examination of the classification performance of the model was made possible by examining the confusion matrix. It offers details on the true positives, true negatives, false positives, false negatives, and insights into particular emotional states that the model finds difficult to discern. Fig. 5 shows the confusion matrix results from the supervised machine-learning models.

It is observed that "Neutral" is often misclassified as "Positive" across multiple models. This misclassification suggests that the models may struggle to differentiate between "Neutral" and "Positive" emotional states, potentially because of the similarities in the underlying neural patterns or features associated with these classes. Similarly, "Negative" is frequently misclassified as "Positive". This misclassification could stem from overlaps in the neural activations or feature representations between "Negative" and "Positive" emotions, making them challenging to distinguish. "Positive" emotions are also prone to misclassification as "Neutral" or "Negative" by the models. This misclassification may occur due to the complexity and variability of "Positive" emotional states, leading to ambiguity in neural structures or feature representations. EEG data are susceptible to noise, artifacts, and physiological confounding factors that affect emotion classification. Despite preprocessing steps, such as normalization and scaling, residual noise or artifacts could still contribute to misclassification.

4.5. Interpretability

Understanding the relationships between the selected features can be valuable for model interpretation. In Fig. 6, the process of LASSO feature selection is illustrated, where the entire dataset consisting of 2,549 features was utilized. LASSO is a regression analysis technique that performs variable selection and regularization to enhance the predictive accuracy and interpretability of a model. Features with nonzero coefficients were considered relevant or selected by the LASSO algorithm. These selected features are retained for model building, whereas features with zero coefficients are effectively discarded because they are deemed less influential for prediction.

Fig. 7 illustrates the outcome of the feature selection process, where the best 20 features were chosen from the initial set of 2,549 features. The features selected in Fig. 7 contribute to the interpretability of the model by focusing on the subset of variables with the most significant impact on the target variable.

From the correlation map of the 20 selected features in Fig. 8, eight features are highly positively correlated, and two are highly inversely correlated. Suppose that one of the highly correlated characteristics is redundant, it may be eliminated from the model, as it becomes difficult to discern each feature's independent impact on the target variable. Understanding the EEG channels or features that contribute most significantly to emotion classification can provide insight into the neural basis

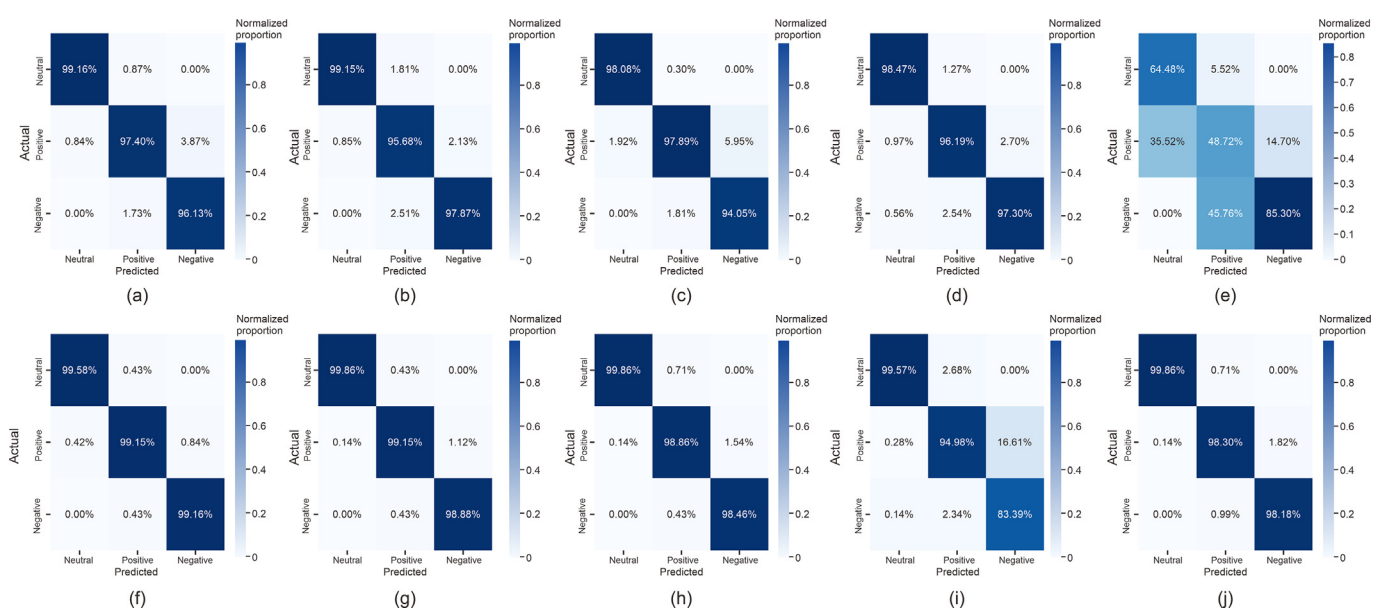

Fig. 5. Confusion matrix results from supervised machine learning models. (a) Confusion matrix of logistic regression classifier; (b) confusion matrix of support vector machine classifier; (c) confusion matrix of k-nearest neighbors classifier; (d) confusion matrix of decision tree classifier; (e) confusion matrix of Gaussian Naive Bayes; (f) confusion matrix of LightGBM classifier; (g) confusion matrix of XGBoost classifier; (h) confusion matrix of CatBoost classifier; (i) confusion matrix of AadBoost classifier; (j) confusion matrix of random forest classifier.

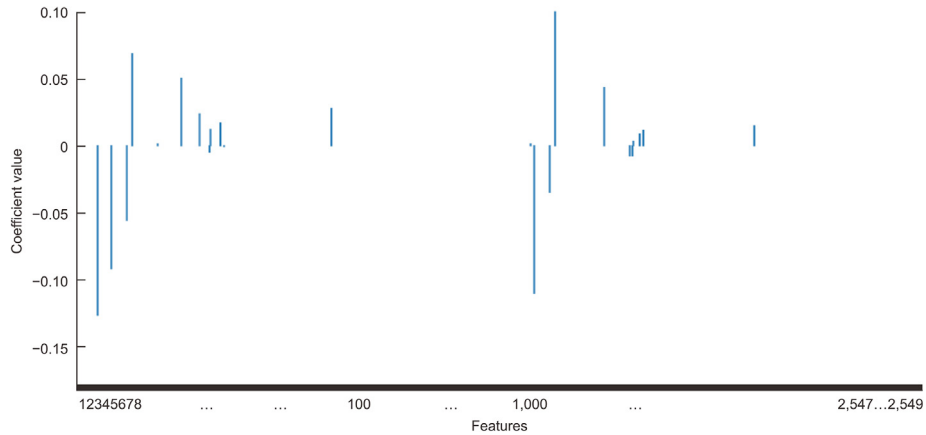

Fig. 6. LASSO feature coefficients for all 2,549 features from electroencephalogram (EEG) brainwave dataset.

of emotions and enhance model interpretability.

In Fig. 8, the positive values indicate a positive correlation (similar activity between channels), while negative values indicate a negative correlation (opposite activity between channels). This correlation does not imply causation, and a correlation map provides insights into the linear relationships between EEG channels. If we have additional information about emotions, we can also explore the relationships between EEG features and emotional states. The correlation map of an EEG dataset is a valuable tool for analyzing the relationships between EEG channels and shedding light on the intricate patterns of neural connectivity in the brain. This is a starting point for further investigations into the neural basis of cognitive processes and emotions, contributing to advancements in neuroscience and brain-computer interface research.

In our analysis, feature selection was implemented to determine which features of the EEG wave dataset affect the emotion classification performance. For tree-based models (such as RF, XGBoost, and LightGBM), the feature values represent the relative contribution of each feature to the model performance improvement at each decision tree split. In linear models such as LR, these values represent the magnitude of the coefficients assigned to each feature, indicating their impact on predicting outcomes. GNB and KNN indirectly assess feature importance by observing the effect on model predictions when values are disturbed. The permutation feature importance values represent the decrease in model performance when the values of a particular feature are perturbed. Higher values suggest greater importance as they lead to a more significant drop in performance. Table 7 summarizes the feature selection and classification performance of various supervised machine learning algorithms.

The observed changes in accuracy following training with the selected 20 features (Table 7) highlight the significant impact of feature selection on model performance. By focusing on a subset of the most relevant features, models may experience varying degrees of improvement or deterioration in accuracy compared with using the entire feature set. While most models experienced a decrease in accuracy, ranging from 5% to 10% after training with the selected features, this trend was consistent across multiple algorithms, including LightGBM, XGBoost, CatBoost, RF, LR, SVM, DTs, and KNN. This suggests that reducing the feature space may have led to a loss of information crucial for the predictive performance of these models. A notable exception is AdaBoost, which exhibits an approximate 20% decrease in accuracy after training with the selected features. This drastic decline suggests that AdaBoost relies heavily on the features excluded during feature selection, leading to a substantial decrease in predictive performance. In contrast, GNB demonstrated an increase in accuracy of approximately 6% after training with the selected features. This improvement suggests that the chosen features contain information that aligns well with the underlying assumptions of the Naive Bayes algorithm, leading to enhanced predictive performance. The results reiterate the importance of careful feature selection tailored to the characteristics and requirements of each model.

Across all supervised machine-learning models, training was notably reduced after feature selection, as shown in Fig. 9. This reduction ranges

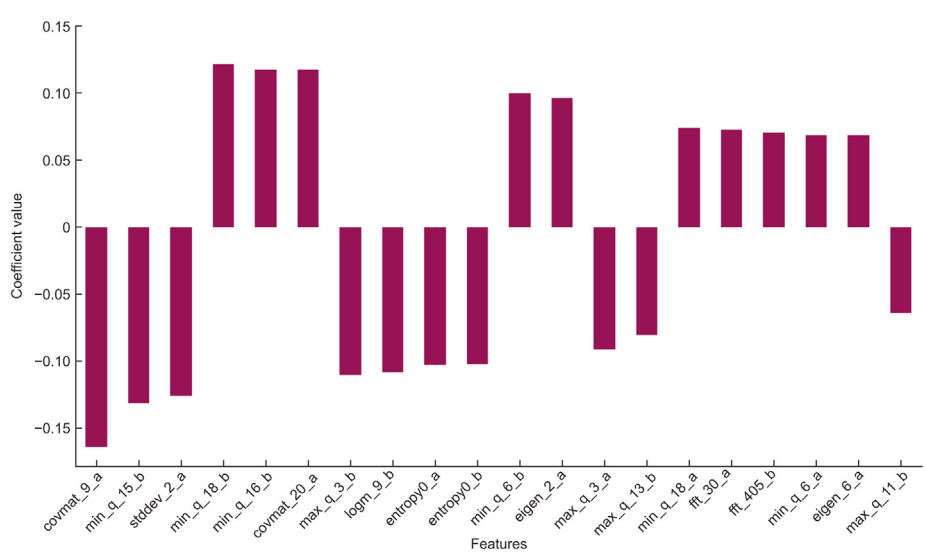

Fig. 7. Selected 20 LASSO feature coefficients.

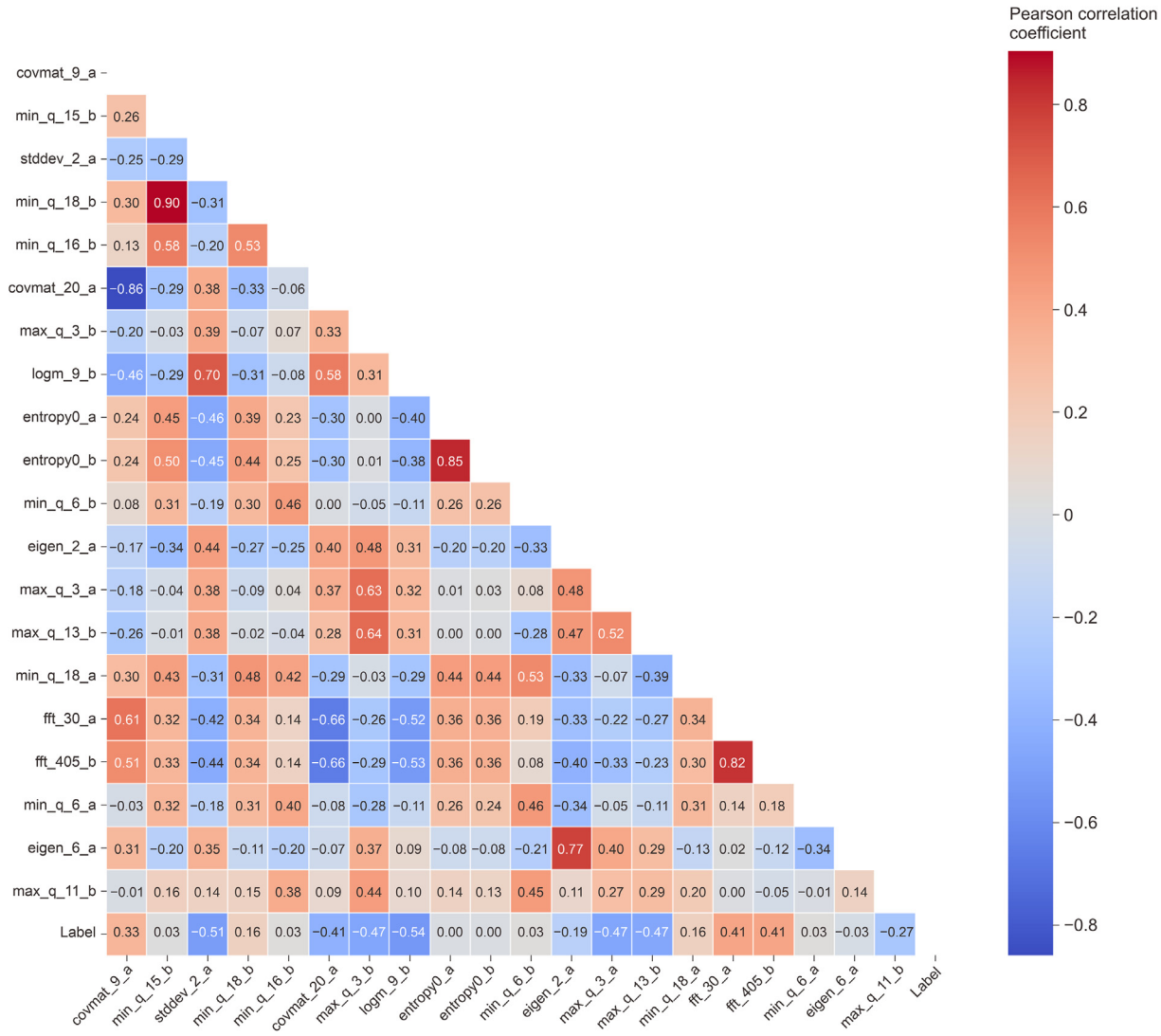

Fig. 8. Correlation matrix of selected 20 features.

Table 7
Feature selection and classification performances of supervised machine-learning algorithms.

Model	Without feature selection					With feature selection				
	Accuracy (%)	Precision (%)	F1-score (%)	Recall (%)	Training time (s)	Accuracy (%)	Precision (%)	F1-score (%)	Recall (%)	Training time (s)
LightGBM	99.30	99.30	99.30	99.30	494.64	95.83	95.83	95.83	95.81	3.90
XGBoost	98.89	98.89	98.89	98.89	497.29	95.97	95.97	95.97	95.95	3.86
CatBoost	98.77	98.77	98.77	98.77	1,673.95	95.40	95.43	95.40	95.37	10.25
Random forest (RF)	98.59	98.42	98.42	98.42	35.59	94.61	94.72	94.61	94.56	4.61
logistic regression (LR)	97.61	97.76	97.75	97.75	27.2	87.95	88.30	87.95	87.63	0.60
Support vector machine (SVM)	97.56	97.58	97.56	97.57	33.36	93.29	93.29	93.29	93.25	0.75
Decision tree (DT)	97.19	97.37	97.37	97.37	32.42	90.38	90.37	90.38	90.38	0.34
k-nearest neighbor (KNN)	96.62	96.68	96.62	96.60	26.83	92.03	92.00	92.03	91.92	0.91
AdaBoost	92.26	92.85	92.26	92.21	234.87	73.03	77.80	73.03	68.04	2.66
Gaussian Naive Bayes (GNB)	66.18	66.16	66.18	64.39	1.77	72.56	77.98	72.56	66.86	0.03

from a few seconds to several minutes, depending on the complexity of the model and the computational resources required for training. CatBoost, XGBoost, and LightGBM exhibit the most significant reductions in computational time after feature selection. CatBoost's training time decreased by approximately 1,663.7 s, while XGBoost's and LightBGM's training time decreased by approximately 493.43 s and 490.74 s, respectively. These substantial reductions suggest that the feature selection has a significant effect on the efficiency of these algorithms. The models RF, LR, SVM, DT, GNB, and KNN experienced notable decreases in computational time after feature selection.

4.6. AUC-ROC curve visualizations

The AUC-ROC curve visualization facilitates a clearer understanding of the model behavior. Fig. 10 shows the AUC-ROC curves of the supervised machine-learning models. The ROC curve is a graphical plot

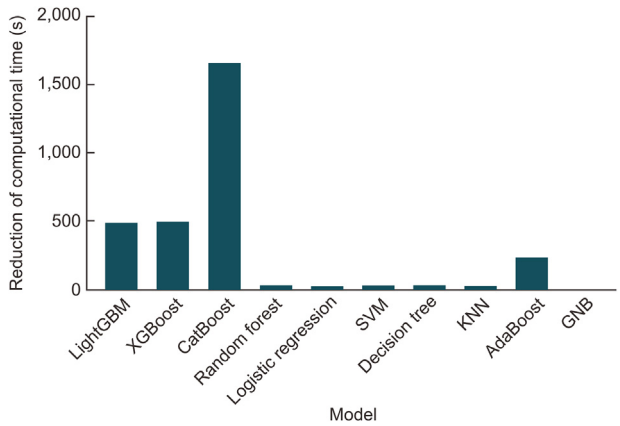

Fig. 9. Analysis of the reduction in computational time after feature selection. SVM: support vector machine; KNN: k-nearest neighbor; GNB: Gaussian Naive Bayes.

illustrating the diagnostic ability of the classification model. AUC of 1 indicated perfect discrimination (i.e., the model made no false-positive or false-negative predictions). The diagonal lines represent the performance of the random classifier. A better-performing classifier should have its ROC curve above this line toward the top-left corner of the plot.

The ROC curves shown in Fig. 10 indicate that the "Neutral" class of emotions was misclassified the most than the "Positive" and "Negative" classes. This could be because features associated with the "Neutral" class might have characteristics that overlap with those of the "Positive" and "Negative" classes, making it challenging for the model to distinguish between the two. The ROC curve analysis reveals that "Neutral" is the class most frequently misclassified by all models. This suggests that distinguishing neutral emotional states from other emotional states poses a significant challenge for the models, potentially because of the unique characteristics of the neutral emotional state or the presence of confounding factors that influence classification decisions.

4.7. Dimensionality reduction techniques

EEG datasets are often subjected to dimensionality reduction techniques to address their high-dimensional characteristics. These methods help minimize the features without sacrificing significant information. The EEG brainwave dataset used in this study contained complex, nonlinear patterns, as is evident from the visualization in Fig. 11. When dimension reduction techniques are used, it is critical to consider the analytical objectives and unique properties of EEG datasets. Furthermore, whether the study is exploratory or a component of a supervised learning task may influence the technique selection. Experimenting with different approaches and assessing their effects on tasks or the interpretability of outcomes is frequently advantageous.

We have implemented different dimension reduction techniques on the EEG brainwave dataset to analyze the impact of its performance on emotion recognition. The performances of the different supervised machine learning models were assessed using the accuracy-recall-precision method, as shown in Table 8.

The analysis of the classification performance for EEG-based emotion recognition using different machine learning models with dimension reduction techniques (PCA, *t*-SNE, and LDA) highlighted that PCA yielded the highest overall accuracy and consistency across models, particularly for Light-GBM, XGBoost, and RF, each achieving approximately 92% accuracy. In contrast, *t*-SNE significantly reduced classification effectiveness, with the top-performing models in PCA showing notably lower accuracy, indicating that *t*-SNE may not be suitable for maintaining separability in this classification context. LDA provided reasonable performance for models such as RF and KNN, achieving accuracies of approximately 81% and 79%, respectively, but failed to match the performance observed with PCA. These findings suggest that PCA is the preferred dimension-reduction technique for this dataset. Simultaneously, LDA could be a secondary option, with *t*-SNE being less practical for optimizing classification in supervised tasks.

5. Limitations and future research directions

Understanding and addressing the limitations of EEG-based emotion recognition are crucial for advancing the field. This section focuses on the limitations that emerged directly from our experiments and grounds the discussion on the practical challenges encountered.

Individual variability introduces significant challenges, as EEG signals vary widely among individuals owing to differences in scalp conductivity, electrode placement, and brain structure, complicating model

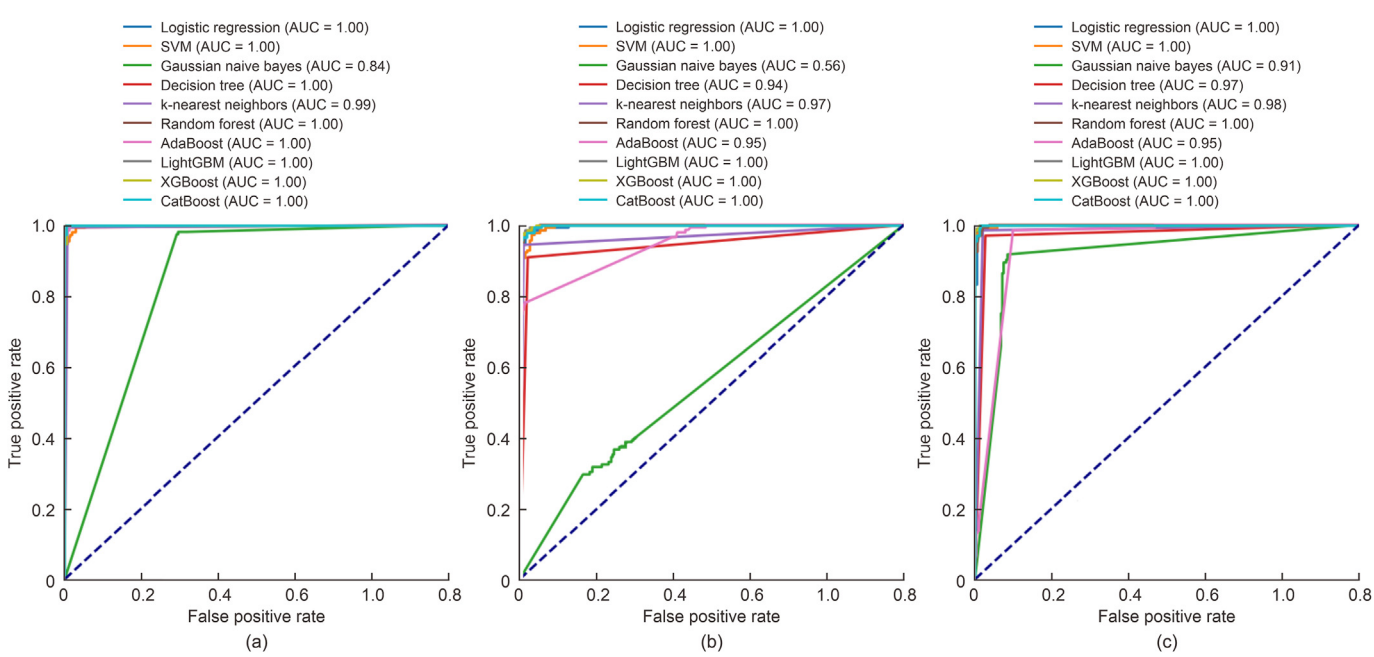

Fig. 10. Area under the receiver operating characteristic (AUC-ROC) curve visualizations of supervised machine learning models. (a) ROC curves for positive class; (b) ROC curves for neutral class; (c) ROC curves for negative class. SVM: support vector machine.

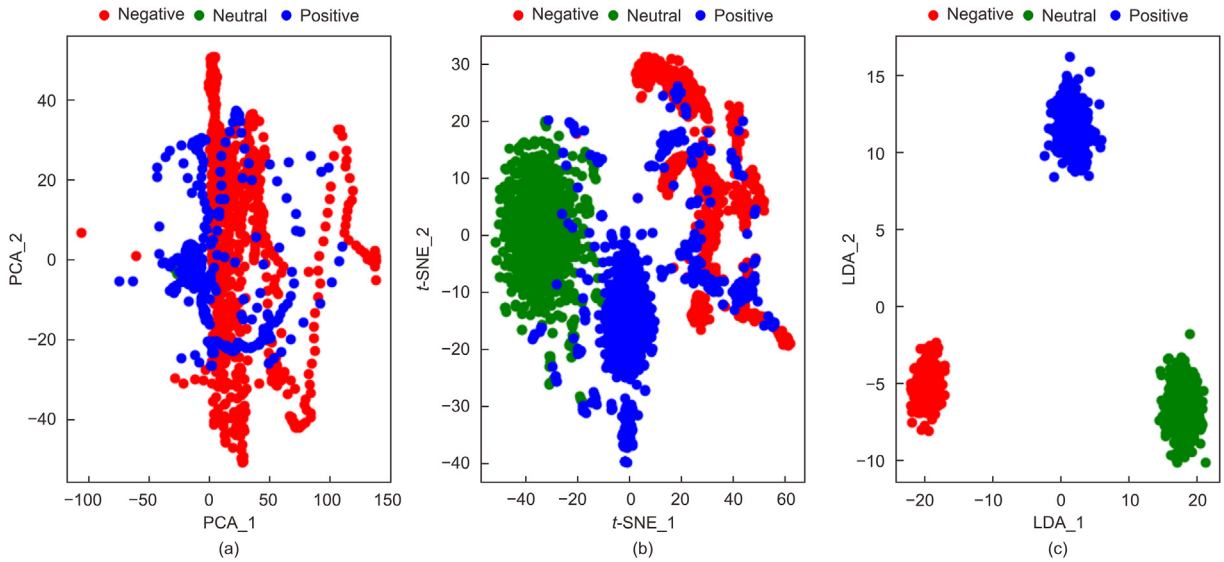

Fig. 11. Dimension reduction techniques applied to the electroencephalogram (EEG) brainwave dataset. (a) principal component analysis (PCA); (b) *t*-stochastic neighbor embedding (*t*-SNE); (C) linear discriminant analysis (LDA).

generalization. The dataset's reliance on only two highly educated STEM students has notable limitations, although it offers specific advantages. The small, homogeneous sample restricts generalizability, as the findings may not extend to broader demographics with different educational backgrounds, ages, or cultures. With limited diversity, the dataset may miss the variability in emotional and cognitive responses that are essential for creating broadly applicable emotion recognition models. Furthermore, models trained on this dataset face a higher risk of overfitting and capturing individual nuances rather than universal patterns. However, the controlled variability of the dataset provides consistency, aiding the early exploration of EEG emotion recognition methods and allowing for in-depth analysis of individual neural responses. Cognitive competence also complicates the analysis because distinguishing between cognitive and emotional EEG signals remains a challenge. Furthermore, context dependency affects generalizability because emotions are influenced by environmental factors that EEG-based models may not fully accommodate, particularly when trained in controlled settings. Additionally, subject cooperation affects the ecological validity of EEG studies because wearing EEG caps can hinder natural emotional expressions.

The difficulty in distinguishing the "Neutral" class from "Positive" and "Negative" emotions reflects a limitation of this study. This challenge arises because neural signals corresponding to the "Neutral" state may share overlapping characteristics with signals from more emotionally charged states. This overlap can blur the distinctions necessary for accurate classification, particularly in models that rely primarily on subtle differences in neural activity. EEG signals are inherently complex and can be influenced by various confounding factors, such as individual differences in baseline neural activity and the subjective nature of emotional states. Consequently, EEG-based models may struggle to consistently and accurately differentiate nuanced or similar emotional states, underscoring the need for improved feature prediction methods and hybrid approaches to refine classification accuracy in emotion recognition tasks.

Another limitation in EEG-based emotion recognition involves computational tradeoffs between different algorithms because distinguishing emotional features often demands complex models that can be resource-intensive. Balancing model accuracy with computational efficiency is essential for improving the classification of subtle emotions without compromising real-time applicability.

Artifact sensitivity presents another limitation. EEG data are highly susceptible to artifacts, such as muscle movements and eye blinks, which require rigorous pre-processing to preserve data quality. Current models often operate on limited emotional dimensions, capturing only primary emotional states, which may not reflect the full spectrum of human emotional complexities. Addressing these limitations requires a multidisciplinary approach, advanced modelling techniques, and a focus on enhancing the applicability of EEGs in real-world scenarios.

Future research should focus on several key areas to address the limitations identified in our study. First, hybrid models can significantly

Table 8
Classification performances of supervised machine-learning algorithms with dimension reduction techniques.

Model	PCA				*t*-SNE				LDA			
	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
LightGBM	92.00	91.78	91.80	91.69	38.00	32.78	37.70	28.07	78.00	78.61	78.22	78.03
XGBoost	92.00	92.27	92.27	92.17	44.00	36.13	44.03	35.56	66.00	69.52	65.57	63.65
CatBoost	91.00	91.00	90.87	90.67	43.00	34.63	43.33	34.95	69.00	70.87	68.62	69.21
Random forest (RF)	92.00	91.57	91.57	91.48	41.00	34.23	40.75	32.98	81.00	80.89	80.80	80.83
Logistic regression (LR)	85.00	86.20	85.48	84.67	56.00	37.82	55.50	44.66	78.00	78.38	77.75	77.59
Support vector machine (SVM)	89.00	89.64	89.46	89.28	35.00	34.27	34.66	26.30	37.00	78.18	36.53	23.09
Decision tree (DT)	89.00	88.68	88.76	88.68	32.00	30.53	32.08	27.76	72.00	72.25	71.90	72.04
k-nearest neighbor (KNN)	91.00	90.59	90.63	90.58	45.00	37.42	44.73	37.86	79.00	79.31	79.39	79.31
AdaBoost	88.00	88.07	87.59	87.14	49.00	39.52	49.18	40.22	79.00	79.29	79.16	78.94
Gaussian Naive Bayes (GNB)	84.00	83.69	84.07	83.80	55.00	37.93	54.80	44.29	73.00	75.32	72.60	71.04

Note: PCA: principal component analysis; SNE: stochastic neighbor embedding; LDA: linear discriminant analysis.

mitigate the shortcomings of individual algorithms. EEG-based models may struggle to consistently and accurately differentiate nuanced or similar emotional states, underscoring the need for improved feature extraction methods and hybrid approaches to refine the classification accuracy in emotion recognition tasks. Additionally, although the dataset served as a suitable starting point for initial testing, future research should include a more diverse participant pool. This enhancement increases the robustness and relevance of the findings across various demographics and emotional expressions. Furthermore, real-time optimization techniques are essential for advancing the practical applications of EEG-based emotion recognition. Investigating neurofeedback as a therapeutic tool for emotion regulation presents an exciting avenue, as it could enable individuals to learn to modulate their emotional responses effectively. Developing advanced deep learning architectures is critical for capturing complex patterns within EEG data and enhancing model performance in real-world scenarios. Transfer learning can also significantly improve model generalization, allowing for more effective emotion recognition in environments with limited labeled data. In addition, longitudinal studies can provide insights into the temporal dynamics of emotional processes, whereas personalized models can enhance recognition accuracy by considering individual differences. Emphasizing ethical considerations in artificial intelligence (AI) development and fostering interdisciplinary collaboration are essential for creating robust, culturally aware, and clinically applicable emotion recognition systems. Collectively, these research directions aim to refine the accuracy, applicability, and understanding of EEG-based emotion-recognition systems.

6. Conclusion

EEG-based emotion recognition in affective computing has evolved from a theoretical idea to a vibrant and exciting field, bridging the gap between neuroscience and technology. EEG-based emotion recognition system provides a unique perspective for understanding the subtleties of emotional states. This has allowed researchers to obtain remarkable insight into the basis of human emotions in the brain. This exploration has revealed a variety of approaches, from the application of cutting-edge machine-learning models to complex signal-processing techniques. These combined endeavors have resulted in notable advancements in comprehending the connection between neural activity and affective states. However, because we reflect on the current state of EEG-based emotion recognition, it is critical to recognize the successes and difficulties ahead. Creating machine-learning models that can identify emotional states with a certain level of accuracy is a noteworthy accomplishment, indicating the possibility of valuable applications in various fields. These ramifications are extensive and include human-computer interaction, healthcare, and education. Recent frameworks have demonstrated that real-time processing capabilities integrate EEG-based emotion identification into interactive systems, creating opportunities for adaptive technologies that react to user emotions in real time.

In this study, we employed the EEG brainwave dataset: The feeling emotions dataset was collected from two subjects to classify their emotions using machine learning and ensemble models. This analysis showed that ensemble models performed better on the EEG brainwave dataset, and CatBoost yielded the best classification results. This observation can be attributed to several factors, such as the complex high-dimensional nature and non-linear data patterns that the ensemble models capture better. Thus, it was experimentally concluded that commercially available EEG headbands are suitable for emotion recognition applications using robust classification algorithms. Intersubject variability remains a significant challenge, highlighting the necessity of customized machine-learning models to account for individual variations in brain activity patterns. Care and caution must be exercised when implementing emotion recognition technologies. Addressing these issues is critical to ensure the dependability, generalizability, and moral application of EEG-based emotion identification algorithms. This field is on the verge of fascinating future opportunities. To achieve more thorough emotion recognition, future research will likely see model improvements, investigate new features, and incorporate multimodal techniques. Furthermore, the fusion of machine learning, neuroscience, and practical applications can lead to creative solutions that affect many aspects of our lives. In conclusion, the path toward EEG-based emotion identification has shed light on the complex relationship between the human mind and technology. The goal of developing technologies that comprehend and react to emotional states with empathy is to drive continued exploration of this frontier in the quest to understand human emotions.

CRediT authorship contribution statement

Anagha Prakash: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Alwin Poulose:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Data availability

The EEG brainwave data that support the findings of this study are openly available in the EEG brainwave dataset: Feeling emotions dataset: https://www.kaggle.com/datasets/birdy654/eeg-brainwave-dataset-feeling-emotions.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly, QuillBot, and ChatGPT in order to improve the grammatical accuracy and clarity of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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References

Ahmad, Z., Khan, N., 2022. A survey on physiological signal-based emotion recognition. Bioengineering 9 (11), 688.
Akhand, M.H., Maria, M.A., Kamal, M.A.S., et al., 2023. Improved EEG-based emotion recognition through information enhancement in connectivity feature map. Sci. Rep. 13 (1), 13804.
Albraikan, A., Tobón, D.P., El Saddik, A., 2019. Toward user-independent emotion recognition using physiological signals. IEEE Sens. J. 19 (19), 8402–8412.
Allen, J.J.B., Reznik, S.J., 2015. Frontal EEG asymmetry as a promising marker of depression vulnerability: summary and methodological considerations. Curr. Opin. Psychol. 4 (Aug.), 93–97.
Alyasseri, Z.A.A., Khader, A.T., Al-Betar, M.A., et al., 2019. EEG signals denoising using optimal wavelet transform hybridized with efficient metaheuristic methods. IEEE Access 8 (Dec.), 10584–10605.
AlZoubi, O., D'Mello, S.K., Calvo, R.A., 2012. Detecting naturalistic expressions of nonbasic affect using physiological signals. IEEE Trans. Affect. Comput. 3 (3), 298–310.
Aqil, M., Jbari, A., Bourouhou, A., 2017. ECG signal denoising by discrete wavelet transform. Int. J. Onl. Eng. 13 (9), 51–68.
Asghar, M.A., Khan, M.J., Rizwan, M., et al., 2020. An innovative multi-model neural network approach for feature selection in emotion recognition using deep feature clustering. Sensors 20 (13), 3765.
Bajaj, N., Requena Carrión, J., 2023. Deep representation of EEG signals using spatio-spectral feature images. Appl. Sci. 13 (17), 9825.

Bao, G., Yan, B., Tong, L., et al., 2021. Data augmentation for EEG-based emotion recognition using generative adversarial networks. Front. Comput. Neurosci. 15 (Dec.), 723843.
Barros, C., Pereira, A.R., Sampaio, A., et al., 2022. Frontal alpha asymmetry and negative mood: a cross-sectional study in older and younger adults. Symmetry 14 (8), 1579.
Bird, J.J., Faria, D.R., Manso, L.J., et al., 2019. A deep evolutionary approach to bioinspired classifier optimisation for brain-machine interaction. Complexity 2019 (1), 4316548.
Bota, P.J., Wang, C., Fred, A.L.N., et al., 2019. A review, current challenges, and future possibilities on emotion recognition using machine learning and physiological signals. IEEE Access 7 (Sep.), 140990–141020.
Britton, J.W., Frey, L.C., Hopp, J.L., et al., 2016. Electroencephalography (EEG): An Introductory Text and Atlas of Normal and Abnormal Findings in Adults, Children, and Infants. American Epilepsy Society, Chicago.
Chaddad, A., Wu, Y., Kateb, R., et al., 2023. Electroencephalography signal processing: a comprehensive review and analysis of methods and techniques. Sensors 23 (14), 6434.
Chakravarthi, B., Ng, S.C., Ezilarasan, M.R., et al., 2022. EEG-based emotion recognition using hybrid CNN and LSTM classification. Front. Comput. Neurosci. 16 (Oct.), 1019776.
Chao, H., Dong, L., 2021. Emotion recognition using three-dimensional feature and convolutional neural network from multichannel EEG signals. IEEE Sens. J. 21 (2), 2024–2034.
Chen, X., Chen, Q., Zhang, Y., et al., 2019. A novel EEMD-CCA approach to removing muscle artifacts for pervasive EEG. IEEE Sens. J. 19 (19), 8420–8431.
Chen, T., Guestrin, C., 2016. XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, pp. 785–794.
Chen, W., Xing, X., Chen, P., et al., 2024. Vesper: a compact and effective pretrained model for speech emotion recognition. IEEE Trans. Affect. Comput. 15 (3), 1711–1724.
Dabas, H., Sethi, C., Dua, C., et al., 2018. Emotion classification using EEG signals. In: Proceedings of the 2018 2nd International Conference on Computer Science and Artificial Intelligence. ACM, pp. 380–384.
Das Chakladar, D., Roy, P.P., Chang, V., 2024. Integrated Spatio-Temporal Deep Clustering (ISTDC) for cognitive workload assessment. Biomed. Signal Process. Contr. 89 (Mar.), 105703.
Dadebayev, D., Goh, W.W., Tan, E.X., 2022. EEG-based emotion recognition: review of commercial EEG devices and machine learning techniques. J. King Saud Univ. Comput. Inf. Sci. 34 (7), 4385–4401.
Dautov, Ç.P., Özerdem, M.S., 2018. Wavelet transform and signal denoising using Wavelet method. In: 2018 26th Signal Processing and Communications Applications Conference (SIU). IEEE, pp. 1–4.
Doma, V., Pirouz, M., 2020. A comparative analysis of machine learning methods for emotion recognition using EEG and peripheral physiological signals. J. Big Data 7 (1), 18.
Duan, R.N., Zhu, J.Y., Lu, B.L., 2013. Differential entropy feature for EEG-based emotion classification. In: 2013 6th International IEEE/EMBS Conference on Neural Engineering (NER). IEEE, pp. 81–84.
Egger, M., Ley, M., Hanke, S., 2019. Emotion recognition from physiological signal analysis: a review. Electron. Notes Theor. Comput. Sci. 343 (May), 35–55.
Gamage, T.A., Kalansooriya, L.P., Sandamali, E.R.C., 2022. An emotion classification model for driver emotion recognition using electroencephalography (EEG). In: 2022 International Research Conference on Smart Computing and Systems Engineering (SCSE). IEEE, pp. 76–82.
Gao, P., Zheng, X., Wang, T., et al., 2024. Graph convolutional neural network based emotion recognition with brain functional connectivity network. Int. J. Crowd Sci. 8 (4), 195–204.
Gao, Q., Wang, C.H., Wang, Z., et al., 2020. EEG based emotion recognition using fusion feature extraction method. Multimed. Tools Appl. 79 (37), 27057–27074.
García-Hernández, R.A., Celaya-Padilla, J.M., Luna-García, H., et al., 2023. Emotional state detection using electroencephalogram signals: a genetic algorithm approach. Appl. Sci. 13 (11), 6394.
Geetha, A.V., Mala, T., Priyanka, D., et al., 2024. Multimodal emotion recognition with deep learning: advancements, challenges, and future directions. Inf. Fusion 105 (May), 102218.
Gong, L., Li, M., Zhang, T., et al., 2023. EEG emotion recognition using attention-based convolutional transformer neural network. Biomed. Signal Process. Contr. 84 (Jul.), 104835.
Gunda, N.K., Khalaf, M.I., Bhatnagar, S., et al., 2024. Lightweight attention mechanisms for EEG emotion recognition for brain computer interface. J. Neurosci. Methods 410 (Oct.), 110223.
Guo, J.Y., Cai, Q., An, J.P., et al., 2022. A transformer based neural network for emotion recognition and visualizations of crucial eeg channels. Phys. A: Stat. Mech. Appl. 603 (Oct.), 127700.
Guo, L., Li, N., Zhang, T., 2024. EEG-based emotion recognition *via* improved evolutionary convolutional neural network. Int. J. Bio Inspired Comput. 23 (4), 203–213.
Habashi, A.G., Azab, A.M., Eldawlatly, S., et al., 2023. Generative adversarial networks in EEG analysis: an overview. J. Neuroeng. Rehabil. 20 (1), 40.
Hu, J., Wang, C., Jia, Q., et al., 2021. ScalingNet: Extracting features from raw EEG data for emotion recognition. Neurocomputing 463 (Nov.), 177–184.
Imani, M., Ali Montazer, G., 2019. A survey of emotion recognition methods with emphasis on E-Learning environments. J. Netw. Comput. Appl. 147 (Dec.), 102423.
Iyer, A., Das, S.S., Teotia, R., et al., 2023. CNN and LSTM based ensemble learning for human emotion recognition using EEG recordings. Multimed. Tools Appl. 82 (4), 4883–4896.
Jain, V., Parab, K., Kalgutkar, S., et al., 2021. EEG brainwave emotion detection using stacked ensembling method. In: 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT). IEEE, pp. 1–7.
Jha, S.K., Suvvari, S., Kumar, M., 2024. Emotion recognition from electroencephalogram (EEG) signals using a multiple column convolutional neural network model. SN Comput. Sci. 5 (2), 213.
Kalashami, M.P., Pedram, M.M., Sadr, H., 2022. EEG feature extraction and data augmentation in emotion recognition. Comput. Intell. Neurosci 2022 (1), 7028517.
Ke, G., Meng, Q., Finley, T., et al., 2017. LightGBM: a highly efficient gradient boosting decision tree. In: 31st International Conference on Neural Information Processing Systems, Long Beach, United States, pp. 3149–3157.
Khan, M., Gueaieb, W., El Saddik, A., et al., 2024. MSER: Multimodal speech emotion recognition using cross-attention with deep fusion. Expert Syst. Appl. 245 (Jul.), 122946.
Khatun, S., Mahajan, R., Morshed, B.I., 2016. Comparative study of wavelet-based unsupervised ocular artifact removal techniques for single-channel EEG data. IEEE J. Transl. Eng. Health Med. 4 (Mar.), 2000108.
Kim, J.H., Poulose, A., Han, D.S., 2021. The extensive usage of the facial image threshing machine for facial emotion recognition performance. Sensors 21 (6), 2026.
Li, H., Qing, C., Xu, X., et al., 2017. A novel DE-PCCM feature for EEG-based emotion recognition. In: 2017 International Conference on Security, Pattern Analysis, and Cybernetics (SPAC). IEEE, pp. 389–393.
Li, Y., Wang, P.T., Vaidya, M.P., et al., 2021. Electromyogram (EMG) removal by adding sources of EMG (ERASE)-a novel ICA-based algorithm for removing myoelectric artifacts from EEG. Front. Neurosci. 14 (Jan.), 597941.
Li, Z., Qiu, L., Li, R., et al., 2020. Enhancing BCI-based emotion recognition using an improved particle swarm optimization for feature selection. Sens. Basel 20 (11), E3028.
Lin, W., Li, C., 2023. Review of studies on emotion recognition and judgment based on physiological signals. Appl. Sci. 13 (4), 2573.
Liu, S., Wang, X., Zhao, L., et al., 2021. Subject-independent emotion recognition of EEG signals based on dynamic empirical convolutional neural network. IEEE/ACM Trans. Comput. Biol. Bioinform. 18 (5), 1710–1721.
Liu, J., Wu, H., Zhang, L., et al., 2022. Spatial-temporal transformers for EEG emotion recognition. In: 2022 The 6th International Conference on Advances in Artificial Intelligence. ACM, pp. 116–120.
Liu, Z.T., Xie, Q., Wu, M., et al., 2018. Electroencephalogram emotion recognition based on empirical mode decomposition and optimal feature selection. IEEE Trans. Cogn. Dev. Syst. 11 (4), 517–526.
Liu, Y.J., Yu, M., Zhao, G., et al., 2018. Real-time movie-induced discrete emotion recognition from EEG signals. IEEE Trans. Affect. Comput. 9 (4), 550–562.
Lövheim, H., 2012. A new three-dimensional model for emotions and monoamine neurotransmitters. Med. Hypotheses 78 (2), 341–348.
Luo, Y., Zhu, L.Z., Wan, Z.Y., et al., 2020. Data augmentation for enhancing EEG-based emotion recognition with deep generative models. J. Neural Eng. 17 (5), 056021.
Ma, W., Zheng, Y., Li, T., et al., 2024. A comprehensive review of deep learning in EEG-based emotion recognition: classifications, trends, and practical implications. Peerj Comput. Sci. 10 (5), e2065.
Maddirala, A.K., Veluvolu, K.C., 2022. ICA with CWT and k-means for eye-blink artifact removal from fewer channel eeg. IEEE Trans. Neural Syst. Rehabil. Eng. 30 (May), 1361–1373.
Masrur Ahmed, S., Tanzim Sabur, E., 2023. Emotion analysis on eeg signal using machine learning and neural network. arXiv:2307.05375.
Mayor Torres, J.M., Clarkson, T., Hauschild, K.M., et al., 2022. Facial emotions are accurately encoded in the neural signal of those with autism spectrum disorder: a deep learning approach. Biol. Psychiatry Cogn. Neurosci. Neuroimaging 7 (7), 688–695.
Mennella, R., Patron, E., Palomba, D., 2017. Frontal alpha asymmetry neurofeedback for the reduction of negative affect and anxiety. Behav. Res. Ther. 92 (May), 32–40.
Miao, M., Hu, W., Xu, B., et al., 2022. Automated CCA-MWF algorithm for unsupervised identification and removal of EOG artifacts from EEG. IEEE J. Biomed. Health Inform. 26 (8), 3607–3617.
Nandini, D., Yadav, J., Rani, A., et al., 2024. Enhancing emotion detection with non-invasive multi-channel EEG and hybrid deep learning architecture. Iran. J. Sci. Technol. Trans. Electr. Eng. 48 (3), 1229–1248.
Nawaz, R., Cheah, K.H., Nisar, H., et al., 2020. Comparison of different feature extraction methods for EEG-based emotion recognition. Biocybern. Biomed. Eng. 40 (3), 910–926.
Negi, T., 2021. Analysis and processing of EEG signal: a review. In: Proceedings of the International Conference on Smart Data Intelligence (ICSMDI 2021).
Pal, S., Mukhopadhyay, S., Suryadevara, N., 2021. Development and progress in sensors and technologies for human emotion recognition. Sensors 21 (16), 5554.
Pane, E.S., Wibawa, A.D., Purnomo, M.H., 2019. Improving the accuracy of EEG emotion recognition by combining valence lateralization and ensemble learning with tuning parameters. Cogn. Process. 20 (4), 405–417.
Petrantonakis, P.C., Hadjileontiadis, L.J., 2015. EEG-based emotion recognition using advanced signal processing techniques. In: Emotion Recognition: A Pattern Analysis Approach. Wiley, Hoboken, pp. 269–293.
Rakibul Mowla, M., Ng, S.C., Zilany, M.S.A., et al., 2015. Artifacts-matched blind source separation and wavelet transform for multichannel EEG denoising. Biomed. Signal Process. Contr. 22 (Sep.), 111–118.
Piho, L., Tjahjadi, T., 2020. A mutual information based adaptive windowing of informative EEG for emotion recognition. IEEE Trans. Affect. Comput. 11 (4), 722–735.

Prokhorenkova, L., Gusev, G., Vorobev, A., et al., 2018. CatBoost: unbiased boosting with categorical features. In: Proceedings of the 32nd International Conference on Neural Information Processing Systems, pp. 6639–6649.
Putra, A.E., Atmaji, C., Ghaleb, F., 2018. EEG-based emotion classification using wavelet decomposition and k-nearest neighbor. In: 2018 4th International Conference on Science and Technology (ICST). IEEE, pp. 1–4.
Qiao, W., Sun, L., Wu, J., et al., 2024. EEG emotion recognition model based on attention and GAN. IEEE Access 12 (Feb.), 32308–32319.
Saganowski, S., Perz, B., Polak, A.G., et al., 2023. Emotion recognition for everyday life using physiological signals from wearables: a systematic literature review. IEEE Trans. Affect. Comput. 14 (3), 1876–1897.
Schapire, R.E., 2013. Explaining AdaBoost. In: Schölkopf, B., Luo, Z., Vovk, V. (Eds.), Empirical Inference. Springer, Berlin, Heidelberg, pp. 37–52.
Sheoran, P., Saini, J.S., 2020. A new method for automatic electrooculogram and eye blink artifacts correction of EEG signals using CCA and NAPCT. Procedia Comput. Sci. 167, 1761–1770.
Shu, L., Xie, J., Yang, M., et al., 2018. A review of emotion recognition using physiological signals. Sens. Basel 18 (7), E2074.
Stajić, T., Jovanović, J., Jovanović, N., et al., 2022. Comparison of machine learning approaches to emotion recognition based on deap database physiological signals. Telfor J. 14 (2), 73–78.
Su, Y., Zhang, Z., Li, X., et al., 2022. The multiscale 3D convolutional network for emotion recognition based on electroencephalogram. Front. Neurosci. 16 (Aug.), 872311.
Tibdewal, M.N., Mahadevappa, M., Ray, A.K., et al., 2016. Power line and ocular artifact denoising from eeg using notch filter and wavelet transform. In: 2016 3rd International Conference on Computing for Sustainable Global Development (INDIACom). IEEE, pp. 1654–1659.
Veeramallu, G.K.P., Anupalli, Y., Kumar Jilumudi, S., et al., 2019. EEG based automatic emotion recognition using emd and random forest classifier. In: 2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT). IEEE, pp. 1–6.
Wang, J., Wang, M., 2021. Review of the emotional feature extraction and classification using EEG signals. Cogn. Robot. 1, 29–40.
Wang, Z., Wang, Y., Hu, C., et al., 2022. Transformers for EEG-based emotion recognition: a hierarchical spatial information learning model. IEEE Sens. J. 22 (5), 4359–4368.
Wei, C., Chen, L.L., Song, Z.Z., et al., 2020. EEG-based emotion recognition using simple recurrent units network and ensemble learning. Biomed. Signal Process. Contr. 58 (Apr.), 101756.
Wei, J., Hu, G., Yang, X., et al., 2024. Learning facial expression and body gesture visual information for video emotion recognition. Expert Syst. Appl. 237 (Mar.), 121419.
Wei, Y., Liu, Y., Li, C., et al., 2023. TC-net: a transformer capsule network for EEG-based emotion recognition. Comput. Biol. Med. 152 (Jan.), 106463.
Yaacob, M., Gunawan, T.S., Abu Bakar, M.I.F., et al., 2024. Accurate EEG-based emotion recognition using LSTM and BiLSTM networks. In: 2024 IEEE 10th International Conference on Smart Instrumentation, Measurement and Applications (ICSIMA). IEEE, pp. 13–18.
Yamin, M.N.M., Aziz, K.A., Siang, T.G., et al., 2023. Exploring the research landscape of automated emotion recognition system adoption in Malaysia: a systematic literature review. J. Advances Artif. Life Robot. 3 (4), 193–204.
Yan, X., Boudrias, M.H., Mitsis, G.D., 2022. Removal of transcranial alternating current stimulation EEG artifacts using blind source separation and wavelets. IEEE Trans. Biomed. Eng. 69 (10), 3183–3192.
Yang, F., Zhao, X., Jiang, W., et al., 2019. Multi-method fusion of cross-subject emotion recognition based on high-dimensional EEG features. Front. Comput. Neurosci. 13 (8), 53.
Zhang, D.X., Wu, X.P., Guo, X.J., 2008. The EEG signal preprocessing based on empirical mode decomposition. In: 2008 2nd International Conference on Bioinformatics and Biomedical Engineering. IEEE, pp. 2131–2134.
Zhang, Y., Ji, X., Zhang, S., 2016. An approach to EEG-based emotion recognition using combined feature extraction method. Neurosci. Lett. 633 (Oct.), 152–157.
Zhao, G., Ge, Y., Shen, B., et al., 2018. Emotion analysis for personality inference from EEG signals. IEEE Trans. Affect. Comput. 9 (3), 362–371.
Zheng, X., Liu, X., Zhang, Y., et al., 2021. A portable hci system oriented eeg feature extraction and channel selection for emotion recognition. Int. J. Intell. Syst. 36, 152–176.
Zhou, W., Gotman, J., 2004. Removal of EMG and ECG artifacts from EEG based on wavelet transform and ICA. In: The 26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE, pp. 392–395.
Zhuang, N., Zeng, Y., Tong, L., et al., 2017. Emotion recognition from EEG signals using multidimensional information in EMD domain. Biomed Res. Int. 2017 (Aug.), 8317357.